

CHAPTER 24

Nutritional Care and Support

KEY TERMS

Anaphylaxis (AN-uh-fuh-LAK-siss)

Anorexia nervosa (AN-uh-REK-see-ah ner-VOH-sah)

Bolus feeding (BOH-lus FEE-ding)

Bulimia nervosa (buh-LEE-mee-ah ner-VOH-sah)

Clear liquid diet (KLEER LIK-wid DYE-uht)

Enteral nutrition (EN-ter-uhl new-TRISH-un)

Food intolerance (FOOD in-TALL-er-ens)

Full liquid diet (FUL LIK-wid DYE-uht)

Gastric decompression (GAS-trik DEE-kom-PRESH-un)

Hemoglobin A_{1c} (HEE-muh-GLOW-bin A_{1c})

Hyperglycemia (HYE-per-glye-SEE-mee-ah)

Hypoglycemia (HYE-poh-glye-SEE-mee-ah)

Jejunostomy tube (JAY-joo-NAWS-toh-mee TOOB)

Mechanical soft diet (muh-KAN-i-kuhl SOFT DYE-uht)

Nasogastric tube (NAY-zoh-GAS-trik TOOB)

NPO

Parenteral nutrition (pah-REN-tur-uhl new-TRISH-un)

Partial parenteral nutrition (PAR-shul pah-REN-tur-uhl new-TRISH-un)

Percutaneous endoscopic gastrostomy tube (PER-kyoo-TAY-nee-us EN-duh-SKOP-ik gas-TRAWS-toh-mee TOOB)

Pureed diet (PYOO-rayd DYE-uht)

Total parenteral nutrition (Toh-tuhl pah-REN-tur-uhl new-TRISH-un)

CHAPTER CONCEPTS

Caring
Nutrition
Safety

LEARNING OUTCOMES

1. Define various terms associated with nutritional care and support.
2. Discuss methods to assist with meals and improve the patient eating experience in the hospital.
3. Explain how to monitor intake and output and why it is important.
4. Describe NPO status, regular diets, and diets modified by consistency.
5. Describe diets modified for diseases and preferences.
6. Describe the nurse's responsibilities for patients on special diets, including diabetics and their specific needs.
7. List at least six guidelines to use in teaching patients with diabetes.
8. Identify common eating disorders.
9. Describe how drugs can affect food intake, absorption, metabolism, and excretion of nutrients.
10. Describe nasogastric and nasointestinal tubes, including size, length, and lumens.
11. Explain when and how nasogastric tubes are used for gastric decompression.
12. Discuss how feeding tubes are placed and used to provide nutrition.
13. Differentiate between intermittent and continuous tube feedings.
14. Describe the process for and importance of checking placement of nasogastric and nasointestinal tubes and checking residual gastric volume.
15. Identify nursing responsibilities for patients with feeding tubes, including the prevention of

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15. Identify nursing responsibilities for patients with feeding tubes, including the prevention of complications.
16. Discuss the use of partial and total parenteral nutrition to maintain nutritional status.
17. Discuss information found in the Connection features in this chapter.
18. Identify specific safety information.
19. Answer questions about the skills in this chapter.

SKILLS

- [24.1 Assisting a Patient With Feeding](#)
- [24.2 Recording Intake and Output](#)
- [24.3 Inserting a Nasogastric Tube](#)
- [24.4 Irrigating a Nasogastric Tube](#)
- [24.5 Removing a Nasogastric Tube](#)
- [24.6 Administering Intermittent Bolus Enteral Feedings](#)
- [24.7 Using a Feeding Pump](#)

CRITICAL THINKING CONNECTION

Clinical Assignment

You are assigned to care for a 58-year-old male patient who has suffered a severe cerebrovascular accident (CVA). He is unable to swallow any food or fluids without choking and has been hydrated with IV fluids for the 2 days since the CVA occurred. The health-care provider has decided that the best course of action is to place a nasogastric (NG) tube and provide tube feedings for the next few weeks. After that, the patient may be able to regain his ability to swallow and can begin to eat soft foods again.

Critical Thinking Questions

You will most likely be placing the NG tube for this patient.

1. What do you need to know before you perform the procedure?
2. What complications could occur?
3. How will you take measures to prevent them?

As you provide care to patients, you will encounter those who require nutritional support in the form of modified diets, nutritional restrictions, supplementation, or mechanical assistance with nutrition. In this chapter, you will learn about the nutritional needs of patients and different types of therapeutic diets, including diets modified for disease states. You will learn to assess a patient's ability to eat or need for alternative means of nutrition and how to assist a patient with eating and how to make mealtimes pleasant. You will also learn to care for patients receiving enteral and parenteral nutrition, patients with feeding tubes, the different types of feeding formulas, and the care and management of feeding tubes.

SUPPORTING NUTRITIONAL INTAKE

Hospitalized patients may experience a variety of factors related to illness or medication that impede their nutritional intake, such as anxiety, pain, fatigue, lack of appetite (called *anorexia*), or nausea and vomiting. Your role as the nurse in nutritional care is to be supportive and ensure that the patient has adequate nutrient intake. This is more than just ensuring that the patient's meal is ordered or that the meal tray is delivered to the correct patient. Food and mealtimes in our society are often associated with social events or cultural or ethnic traditions, and most people have their own preferred routine for meals. For those patients receiving care in an institutional setting, individual patient needs may become secondary to the needs of the organization. Patient meal choices often are limited, and mealtimes are dictated by the needs of the various organizational services. Nursing intervention can improve the patient experience and optimize nutritional intake.

Assessment

Assessment

When a patient is admitted to the hospital, it is important to assess their nutritional problems and needs. It is impossible to meet a patient's specific needs if you do not know what the patient's needs are. During the initial assessment, you should ask questions to identify specific allergies, special diets the patient may have been following at home, and dietary preferences the patient may have, including cultural or ethnic requirements or restrictions. You will also want to assess the patient's physical capabilities and identify whether assistance with feeding is required. Interventions may include cutting up foods, assisting in opening containers, or feeding the patient who is blind or too weak to do so independently.

Another assessment you will perform is to determine what medications the patient has been taking at home because many medications can create side effects such as nausea, diminished appetite, or altered sense of taste. It often is possible to assist the patient to overcome medication side effects with simple interventions. These may include helping the patient to select foods or arranging meal and snack times that minimize nausea, thus helping to improve the patient's food intake.

Mealtime Preparation

It is sad to have to write this, but because it does happen, we feel that we should point it out. Sometimes "passing the meal trays" is treated as just that, a task to be completed and nothing more. It is not appropriate to walk down a hospital hallway and see meal trays sitting on the overbed table next to the bed, with dish covers still in place and no one eating the meal. This is a problem because it is difficult for the ill body to heal without adequate fluids and nutrition. It is your job as the nurse to do everything in your power to ensure all patients are prepared for each meal and are assisted as needed throughout the meal ... not just to set the meal tray down and leave the room.

Odors

It is important to provide patients with a pleasant environment that is free of clutter and odors to enhance mealtimes. Odors can adversely affect the patient's desire to eat or may trigger nausea. Work to eliminate body excretion smells such as feces and infectious or bloody drainage. It is important for you to guard against breath odor (known as *halitosis*), body odor, and food odor on your hands, all of which can be a source of nausea, headache, or allergic reaction for the patient. Take care to eliminate these odors if they are present before mealtime.

Ill patients often are negatively affected by odors that normally would be pleasant or, at the very least, not offensive, such as perfume or aftershave, fragranced hair products, lotions, and strong fabric softeners. Avoid all personal care products that contain fragrance and laundry products that result in a strong residual fragrance on your clothing.

Environment

Plan to prepare the patient's room before the meal trays are delivered so that the overbed table on which you will set the tray is ready. Make certain that the table is free of offensive items such as an emesis basin containing sputum, a urinal with or without urine in it, a roll of toilet tissue, a hair brush, or soiled tissues or washcloths. Remove anything that you would not want on your dining room table during mealtime. Wipe the table surface with a damp cloth or disinfectant if contaminated items have been on it.

Toileting

Inquire whether or not the patient needs to go to the bathroom, and, if so, assist the patient to the bathroom or with use of a bedpan. A patient who has a full bladder or needs to have a bowel movement probably will not eat as much as they would after toileting. If a patient has to interrupt a meal to attend to toileting, there is little likelihood that the patient will resume eating afterward.

Patient Comfort

Make certain the patient is comfortable because nausea, pain, or discomfort can inhibit eating. Make certain that the patient has slippers or socks on to keep the feet warm and either a robe or a second gown worn backward, with the opening in front similar to a robe. The patient may also need a blanket placed across the lap if they feel cold. If the patient wears dentures or glasses, obtain them and make sure that they are clean. Then assist the patient as needed to insert the dentures

them and make sure that they are clean. Then assist the patient as needed to insert the dentures or put on the glasses.

Patient Positioning

If the patient is able to sit on the side of the bed or get up to sit in a chair, assist the patient to do so. An upright position is more conducive to eating. If not, make certain that the patient is positioned properly in the bed with good body alignment and then elevate the head of the bed between 60 and 90 degrees depending on the patient's condition and preference. Position the overbed table over the patient's lap so that it is directly in front of the patient ([Fig. 24.1](#)). Adjust the height of the table according to the patient's needs.



FIGURE 24.1 When feeding a patient, position the overbed table in front of the patient, not off to the patient's side at the side of the bed.

Patient Cleanliness

To reduce the risk of bacterial contamination while eating, provide the patient with two warm wet washcloths (one with soap and one without soap for rinsing) so they can wash their face and hands before eating or a rinse-free bath wipe. If the patient is unable to perform this task independently, assist as needed. Also, make certain that you wash your hands before delivering and setting up the meal tray.

KNOWLEDGE CONNECTION

Explain the importance of preparing the patient and the patient's environment before delivering meal trays. What specific steps should be taken to prepare the patient before meal trays arrive? Discuss patient comfort in relation to mealtimes. What are your responsibilities regarding cleanliness and sanitation for mealtimes? What is included in preparing the overbed table to receive the meal tray?

Assistance With Eating

Your time spent assisting patients with meals can be used to learn more about your patients and to develop and enhance the nurse-patient relationship. Patients who require assistance often find themselves in a position that they are not used to, that of dependency on others to help them meet their needs. During mealtime, take time to talk with your patient and ask questions about their life situations and experiences. Probe to determine whether the patient has anxiety, concerns, or questions that have not been answered or to gather insight into the dynamics of their life. Help to

make mealtime more of a social event or at least a more pleasant interaction for the patient unless the patient's culture views it differently. Mealtimes also offer an opportune time to educate patients regarding dietary changes they may need to make or the use of adaptive equipment. [Box 24.1](#) offers tips on ways to assist patients with meal planning and eating. For more information regarding the steps of assisting a patient with feeding, see [Skill 24.1](#).

Assisting a patient with meals is a task that can be delegated to assistive personnel or certified nursing assistants (CNAs). However, before delegating the task, make certain that the assistive personnel or CNA understands the ethnic and cultural considerations related to different foods, nutrition, and mealtime. You may need to educate the assistive personnel or CNA regarding ethnic and cultural beliefs and customs related to food and its consumption. It is also your responsibility to ensure that the assistive personnel or CNA is knowledgeable about the patient's physiological status as it relates to impaired swallowing, which would require the patient to be fed slowly. They must know what to do if the patient chokes or aspirates.

Box 24.1

Tips for Assisting Patients With Meal Planning and Eating

Because good nutrition is so important to the healing process, it must be monitored and evaluated for adequacy. The patient commonly does not feel like eating much, so it is your responsibility to make it a priority to do everything you can to ensure optimal nutritional intake. Following are a few tips to help you to do this:

- Spend time getting to know the patient. Be warm and compassionate, letting the patient know you understand that their pain, anxiety, or other psychosocial factors are affecting appetite.
- If the patient is nauseated or hurting, provide appropriate medication before mealtime. You cannot expect a patient who is nauseated or having pain to eat adequately to meet their nutritional needs.
- Make certain the patient is prepared for mealtime by getting the patient comfortable in a bed or chair, providing a warm wet washcloth to wash the hands and face (one with soap and one without soap for rinsing) or rinse-free bath wipe, and clearing the overbed table of clutter and unpleasantities to provide ample room for the meal tray. Think of how you would prepare your kitchen table for a meal at home. (Would you want to eat lunch with a roll of toilet tissue, a urinal, or an emesis basin containing green sputum next to your plate?) **Safety: Wipe off the overbed table with disinfectant if a urinal or other contaminated items have lain on it.** This preparation should be carried out before tray arrival because it is impossible to do these tasks with a tray of food in one's hands.
- You and any assistive personnel should always wash your hands before passing out meal trays.
- Ensure that each meal tray is assessed for the correct diet and appropriate temperature of food.
- Many patients are older or frail; have poor eyesight or weakness because of illness or stroke; or have limited use of extremities related to IVs, dressings, and casts. Open all containers on the tray, such as milk cartons, juice containers, cellophane packaging for plastic utensils, and condiments. Many patients who are unable to accomplish these seemingly simple tasks will just not eat rather than have to ask for assistance.
- Take your time while feeding patients. Avoid rushing the patient.
- Be certain that assistive personnel know how to safely assist those patients who require assistance.
- Help the patient identify factors that would improve the eating experience, such as selecting foods they like, creating a comfortable environment, arranging for early or later trays if the patient's normal routine is quite different from the hospital routine, or allowing friends and family to bring in foods from home provided there are no dietary restrictions.
- If the patient's appetite is limited, provide five to six smaller feedings per day rather than three larger ones. The health-care provider will generally write an order for this if you ask.
- Educate your patient as to the importance of nutritional intake and drinking adequate fluids in the healing process.
- Make rounds to your patients' rooms during mealtimes. Ask your patients if they have any

- Make rounds to your patients' rooms during mealtimes. Ask your patients if they have any needs or if they are satisfied with their meals; help facilitate a pleasant experience by being a problem-solver.

Monitoring Intake and Output

Because fluid and electrolyte balance is critical to optimal health and recovery from illness and injury, ensuring adequate fluid intake is even more important than ensuring consumption of food. This should be a primary focus in your care of each patient. When there are concerns regarding a patient's fluid and electrolyte balance, it will be your responsibility to monitor the patient's intake and output, commonly called I&O. This simply means to measure and record all fluids taken in and all fluid volumes that are lost.

Intake includes fluids taken by mouth, those administered IV, and fluids administered by enteral or parenteral feedings, which will be discussed later in this chapter. A clear plastic drinking cup marked in increments of ounces and milliliters can be used to measure the amount of fluid that is drunk. It is also used to measure the remaining volume after the patient drinks only part of a container with a specified volume in it. For example, if a can of carbonated cola contained 360 mL and the patient drank only part of the cola, you would measure the remainder and subtract that from the 360 mL to determine the amount ingested. The cup should be held at approximately eye level to accurately read the volume ([Fig. 24.2](#)). Most hospitals have standardized volumes listed for their commonly used containers, making it easier to calculate intake. [Box 24.2](#) provides a list of common container volumes.

Output includes all bodily fluids that are lost, including urine, emesis, liquid stool, blood, suctioned gastric contents, and drainage from drainage devices such as a Hemovac or Jackson-Pratt drain, which you will read about in [Chapter 26](#). You will measure urine and emesis in clear containers that are larger than the one used to measure oral intake. The units of measurement will be the same: ounces and milliliters. You may hear the containers referred to as graduated containers or receptacles. When you measure any form of output, you will write it on the I&O sheet at the bedside, indicating the type of output, such as Hemovac drainage or gastric suction. Each category is totaled separately and documented on the I&O sheet or flow sheet in the patient's chart at the end of each shift. Then the total of all intake is calculated and compared with the calculated total of all output. The average adult requires between 1,500 and 2,500 mL of fluid intake per 24 hours depending on the patient's size and health status. The total urinary output should closely balance with the total intake, usually within 200 to 300 mL of the intake. [Skill 24.2](#) provides additional information regarding intake and output recording.



FIGURE 24.2 Use a graduated cup and hold up at eye level to accurately measure intake.

Box 24.2

Examples of Standard Container Volumes

Different facilities will use glasses, cups, and bowls of various sizes that will hold differing volumes of liquid. Each facility will have its own list of standard container volumes. Become familiar with the volumes where you work. The following is a list of common container volumes:

the volumes where you work. The following is a list of common container volumes:

- **Popsicle:** 90 mL
- **Juice glass:** 120 mL
- **Individual ice cream cup:** 120 mL
- **Coffee cup:** 150 mL
- **Small gelatin cup:** 180 mL
- **Drinking glass:** 180 mL or 240 mL
- **Small soup bowl:** 180 mL
- **Large soup bowl:** 240 mL
- **Small carton of milk:** 240 mL

KNOWLEDGE CONNECTION

Describe how you can make mealtime more conducive to eating and more pleasant for the patient. What can you do to help increase nutritional intake for the patient with a limited appetite? Discuss at least five ways you can assist patients with meal planning and eating. Why is ensuring intake and output balance pertinent? How much fluid should the average adult take in over 24 hours?

Supporting Patients With Special Nutritional Needs

As you provide nursing care and nutritional support to patients, you will encounter patients who have special nutritional requirements. These patients may have an inability to tolerate certain nutrients or may have an actual eating disorder. Food allergies or intolerances present significant challenges. Food allergies are most commonly identified in infancy or early childhood. When identified in children, the patient and family have not had time to develop adaptive measures. Frequently, they are anxious and uncertain about their ability to effectively manage and meet their nutritional needs.

Food Allergies

A *food allergy* is a reaction by the patient's immune system to an allergen, which is a food protein that, once ingested, causes the immune system to develop antibodies. The following are the most common food allergies:

- Peanuts
- Wheat
- Dairy products
- Eggs

Typically, a patient with food allergy experiences symptoms such as skin, respiratory, and gastrointestinal (GI) reactions that get increasingly worse with repeated exposures to the offending agent. It is important that the nurse gather assessment data on the history of the allergy, when it was first identified, symptoms experienced, and interventions used in the event of a reaction. Allergic reactions can present as a rash or hives, or they can be severe and result in anaphylaxis, the most severe form of allergic reaction. **Anaphylaxis** is a life-threatening emergency; it involves swelling of the upper respiratory tract that can result in occlusion of the airway. The patient, the patient's family, and all persons involved with the care of the patient need to be aware of the signs and symptoms of anaphylaxis and the appropriate and immediate treatment of the reaction.

• WORD • BUILDING •

anaphylaxis: ana - against + phylaxis - protection

Food Intolerances

Food intolerance is not an allergic reaction; it is an adverse reaction to a food without activation of the immune response. The symptoms generated by the food intolerance can be triggered by distinct properties of the food, such as tyramine in aged cheese or wine, or by metabolic disorders, such as gluten intolerance. Food intolerance can stem from the body's lack of digestive enzymes or an inability to use these enzymes. The resultant symptoms of intolerance are usually of a GI nature, such as bloating, flatulence, diarrhea, or nausea. Treatment of food intolerances requires

such as bloating, flatulence, diarrhea, or nausea. Treatment of food intolerances requires identification of the offending agent; this can be accomplished by eliminating various foods from the diet over time. Once the offending agent is identified, the patient will need to modify their diet to limit or eliminate the offending foods. Lactose intolerance, for example, would require that a patient limit milk intake. However, the patient also would have to evaluate other foods to determine whether they contain lactose.

KNOWLEDGE CONNECTION

Discuss the differences between food allergy and food intolerance. Describe common symptoms of each.

THERAPEUTIC DIETS

Many patients must follow a modified diet to help manage their illness or injury. Hospital food comes in a variety of regular and modified diets. Diets may be modified singularly, such as a 1,500-calorie diet, or may be modified in combinations, such as a 1,500-calorie, low-sodium, mechanical soft diet. Each facility will have a diet manual that contains details about each type of diet and the foods that are allowed. All patients must have a diet ordered by their health-care provider unless there is an order for the patient to remain NPO, which is explained later. Dietary modifications are made in collaboration with a hospital or institutional dietitian. The following represent the most commonly prescribed restrictions and modified diets.

NPO

NPO stands for “nothing per os,” or mouth. It means the patient is restricted from oral intake of food or fluid, including water. This is frequently ordered before surgery, invasive procedures, and certain diagnostic studies to reduce the risk of aspiration. Common orders may be “NPO after midnight” or “NPO 8 hours before procedure.” A patient who is vomiting may have an NPO order specified to be in effect until nausea and vomiting are no longer a problem.

Most patients who are nourished can tolerate brief periods without food or fluid; however, prolonged periods of NPO status, as when a procedure or surgery is delayed for 24 hours, should be avoided. During prolonged periods of NPO status, IV fluids may be given to maintain hydration and a minimally supportive caloric intake.

Any time a patient is NPO, the patient and family should be informed regarding the NPO status and the reason why it is ordered. This will help to increase compliance. An NPO sign should be placed on the door and on the wall over the patient's bed to serve as a reminder for staff and visitors. Additionally, NPO status should be documented in the patient's electronic health record. You should empty and remove the patient's water carafe and drinking glass from the bedside as a reminder that the patient is not to drink anything.

Regular Diet

A regular diet is appropriate for patients without special nutritional needs. This diet provides approximately 2,000 calories per day and offers a balanced meal plan. The nutritional value of this diet can vary widely depending on the food choices of the patient. Hospital meals offer the patient the opportunity to have adequate caloric intake but are not necessarily designed to be low in sodium, fat, or cholesterol. You may need to educate your patient to make nutritionally sound choices. Regular diets are adapted to meet the age-specific needs of patients, as the nutritional needs of a child differ from those of an older adult and from those of a middle-aged adult.

For patients with specific cultural or ethnic needs, many facilities provide options such as a kosher or vegetarian diet. If a facility does not provide these variations, you may want to encourage the patient's family or friends to bring foods from home that will allow or encourage the patient to eat more.

Even though patients typically order their daily meals from a selection of menu choices, you may encounter patients with complaints related to the taste or texture of hospital food. They may complain of it being “bland and tasteless.” This may be true because hospital diets must be prepared to meet the needs and tastes of a wide variety of patients. Usually, foods available for selection on the hospital menu are limited to avoid unpopular food items such as turnips or those that are fatty, fried, or gas producing.

that are fatty, fried, or gas producing.

Diets Modified by Consistency

Patients with acute illnesses and those undergoing surgery or bowel procedures may need a diet that is modified in consistency to assist in the management of their illness. The health-care provider may order “Clear liquids, advance as tolerated.” This means the patient is allowed to gradually progress from a diet of clear liquids to full liquids, then on to a soft diet, and finally to a regular diet as the patient is able to tolerate the increasing consistencies without nausea, vomiting, or distress.

For some older patients or those with chewing problems or dysphagia (difficulty swallowing), a long-term diet modification may be necessary. Common diets with consistency modifications include clear liquid, full liquid, mechanical soft, and pureed.

- A **clear liquid diet** is ordered to provide hydration and calories in the form of simple carbohydrates that help meet some of the body’s energy needs. The defining characteristic that separates a clear liquid diet from a full liquid diet has to do with its clarity. Is the liquid, either as is or diluted with water, clear enough that you can f“read a newspaper” through it? For example, a dark carbonated cola that is diluted with water would become clear enough to see through it. If so, it may be considered a clear liquid. Examples of clear liquids include water, broth, and tea without milk. As clear liquid diets do not provide adequate calories, proteins, fats, vitamins, or minerals to meet the body’s needs, a patient should be progressed as quickly as possible to a more nutritious diet.
Common examples of when a clear liquid diet might be appropriate include postoperatively or during recovery from GI disease such as vomiting and diarrhea, when the gut must be introduced slowly to food. When the clear liquids are tolerated, the patient may be advanced to a full liquid diet. It will be your responsibility to determine when to advance the diet by assessing how the patient tolerates each new level of diet.
- A **full liquid diet** consists of all the liquids found in a clear liquid diet with the addition of all other opaque liquids and food items that become liquid at room temperature. Examples of full liquids include milk, pudding, and ice cream. As is the case with the clear liquid diet, this diet also provides limited nutrients. Therefore patients should remain on this plan for a limited time. Patients who need to remain on a liquid diet for an extended period should have the diet planned by a dietitian, who may recommend the addition of high-calorie, high-protein supplements.
- A **mechanical soft diet** is the diet of choice for patients with acute or chronic difficulties with chewing, such as those with jaw problems, missing teeth, poorly fitting dentures, or severe weakness or fatigue. This diet includes all of the items from the full liquid diet plus the addition of foods such as scrambled eggs and cottage cheese. Many items can be included in this diet by altering their texture through extensive cooking or use of a blender or food processor. Through the inclusion of a wide variety of foods, this diet is better able to meet a person’s nutritional needs; however, it is quite low in fiber, so patients may be at risk for constipation.
- A **pureed diet** is one that is processed in a blender or food processor. Any food item can be included, but its consistency must be changed. Food items are placed in a blender while liquid is added to moisten them, creating a texture that is somewhat thicker than full liquid but that can still be swallowed easily.

Table 24.1 provides examples of foods for each of the four diets listed earlier.

KNOWLEDGE CONNECTION

What are the categories of diets modified by consistency? Describe the differences between a clear liquid and full liquid diet. Explain what “advance as tolerated” means in relation to a diet order. What assessments should you make before advancing the diet? What are the key nutrients these provide, and which may need to be supplemented?

Table 24.1

Partial List of Foods by Consistency

Clear Liquid	Full Liquid	Mechanical Soft	Pureed
• Water	• All clear liquids	• All liquids and pureed	• All liquids

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• Water	• All clear liquids	• All liquids and pureed foods	• All liquids
• Broth/bouillon	• Cream soups	• All soups	• Scrambled eggs
• Black coffee	• Milk	• Ground/finely diced meats	• Pureed meats
• Tea	• Cream	• Finely diced/flaked fish	• Pureed vegetables
• Carbonated drinks	• Ice cream	• Cottage cheese	• Pureed fruits
• Clear fruit juices:	• Yogurt	• Soft cheeses	• Mashed potatoes
• Apple	• Milkshakes	• Mashed or riced potatoes	• Gravy
• Grape	• Sherbet	• Rice	• Applesauce
• Cranberry	• Custards/puddings	• Oatmeal	• Baby foods
• Gelatin	• Vegetable juices	• Grits	
• Popsicles	• Pureed vegetables	• Pancakes	
• Clear sports drinks:	• All fruit juices	• Soft breads	
• Gatorade	• Protein drinks	• Soda and graham crackers	
• Electrolyte drinks:	• Liquid supplements	• Cooked, soft vegetables	
• Pedialyte	• Refined cooked cereals:	• Cooked or canned fruits	
	• Cream of Wheat	• Bananas	
	• Malt-O-Meal	• Soft pastries	

Diets Modified for Disease

Certain patient conditions may require that the patient's diet be modified to either increase or restrict the intake of certain nutrients. The following will identify the most common of these diets:

- **Diabetic:** Used to manage calorie and carbohydrate intake for patients with diabetes mellitus, primarily those who are insulin dependent.
- **Calorie restricted:** Used for those patients who must lose weight.
- **Sodium restricted:** Used for those patients with hypertension, heart failure, or kidney or liver failure and for those who require help to prevent or correct fluid retention. The degree of restriction is commensurate with disease severity, severity of symptoms, and the medication regimen prescribed. Typically, sodium restrictions range from 2,000 to 3,000 mg per day. While many patients think that simply not adding salt to foods provides adequate sodium restriction, this is not actually the case. Most of the sodium that people ingest comes from processed foods, making it necessary to read labels for sodium content if the patient is sent home on this diet.
- **Fat restricted:** Used for patients who are experiencing problems with fat malabsorption, such as those with disorders affecting the gallbladder, liver, lymphatic system, pancreas, or intestines. It also may be used for those patients who have elevated triglyceride, lipid, or cholesterol levels and for patients who need general weight loss.
- **Fiber restricted:** Used during the acute phase of intestinal disorders when the presence of fiber may exacerbate intestinal pain, produce diarrhea, or cause an intestinal blockage. This diet is often used before intestinal surgery to minimize fecal volume or after surgery to allow the GI system to transition gradually to a regular diet.
- **Renal:** Used to manage or limit fluids and electrolytes for patients with renal insufficiency or disease. The renal diet is complex, and the patient's nutritional needs change over the course of the disease. A dietitian specializing in renal disorders may guide the nutritional therapy.
- **Protein restricted:** Used to manage protein intake for those patients with liver or kidney disease.
- **High calorie, high protein:** Used to increase calorie and protein intake in those patients with increased need related to wound healing, growth promotion, and increasing or maintaining weight. High-fat foods also may be added to increase calories available for energy use.
- **Five to six small, frequent feedings:** A health-care provider may order any type of diet to be provided in several smaller feedings throughout the day rather than the traditional three meals per day. This is used with those patients who have difficulty tolerating larger meals or who simply are unable to eat more than a few bites at a time, in addition to those who need to have food gradually reintroduced. Often older patients are only able to eat small amounts at a time. This makes it difficult for them to obtain the needed nutrients in the traditional three meals per day. By increasing the number of meal or snack times from three to five or six per day, the patient has a better chance of receiving the needed nutrients.

has a better chance of receiving the needed nutrients.

Diets Modified by Preference

Some patients are on certain diets by choice. Often they feel better if they restrict certain items in the usual diet. Alternately, they may feel strongly about the environmental impact of the food chain in the United States and therefore avoid eating foods that have a negative impact.

- **Gluten-free diets:** People with celiac disease must be on a gluten-free diet. However, other people may have a condition known as “nonceliac gluten sensitivity” and also need to avoid gluten. Still others may have a wheat allergy that requires them to avoid gluten-containing foods. Celiac disease is an autoimmune condition, and wheat allergy causes histamine release. Nonceliac gluten sensitivity is not an autoimmune response but has similar symptoms. Often the diagnosis is made if removing gluten from the diet causes symptoms to subside.

Gluten is present in foods like wheat, barley, flour, couscous, tabouli, and malt vinegar. Sometimes certain meat products and even some pharmaceuticals may contain gluten. There are now more “gluten-free” products available for purchase, but they may be actually lower in some nutrients than those that contain gluten. They may be higher in carbohydrates and fat and lower in fiber and iron, while still costing more than their counterparts. Foods such as buckwheat, lentils, garbanzo beans, and tapioca are all acceptable for a patient on a gluten-free diet.

- **Vegan diets:** People may choose to be vegan for a number of reasons. Some feel it is better for the environment, some feel that animals should not be food sources for people, and others may decide it is a healthier choice for them personally. Some studies have shown that eating a vegan diet has improved or reversed some diseases, particularly cardiovascular diseases.

Those on a vegan diet eat plant-based foods only. They avoid all meat, including fish and poultry, and any other animal products or by-products. They do not consume eggs, dairy products, or honey. This diet is rich in fruits and vegetables, leafy greens, whole grains, nuts, seeds, legumes, and tofu (soy). Alternative products are often used to replace dairy, such as almond or soy milk and cashew cheese. This diet tends to be lower in saturated fat and cholesterol than vegetarian diets. Patients who follow a vegan diet may need supplementation for vitamins B₁₂ and D, calcium, and omega-3 fatty acids.

KNOWLEDGE CONNECTION

Give an example of an illness or condition that might necessitate a high-protein diet. Identify one type of diet that might be ordered to help decrease swelling of a patient's hands and legs. For what conditions might the health-care provider order a fat-restricted diet? What is the advantage of five to six small feedings over the routine three meals per day? High-calorie, high-protein diets might be ordered in what situations?

Nursing Responsibilities

You will be responsible for monitoring the diet type, percentage eaten, and whether or not the patient tolerated the meal. When the health-care provider has ordered “clear liquids, advance as tolerated,” such as might be appropriate postoperatively or during recovery from GI illness, it will be your responsibility to determine when to advance the patient's diet. Using the nursing process, you will assess how much the patient eats and record this as a percentage, record how the patient tolerates each meal, and evaluate whether he or she is ready to have the diet advanced to the next level of consistency. Factors indicating that the patient is not yet ready to advance the diet may include any of the following:

- Nausea
- Vomiting
- Abdominal distention
- Cramping
- Absent or hypoactive bowel sounds
- Complaints of indigestion
- Diarrhea

Once you evaluate how the patient is tolerating each level of increased consistency you will

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Once you evaluate how the patient is tolerating each level of increased consistency, you will notify the dietary service to advance the diet to the next level.

Patients With Diabetes

If your patient has diabetes, you will be expected to monitor not only the amount eaten and the patient's tolerance of the meal but most likely the patient's blood glucose level as well. Refer to the Laboratory and Diagnostic Connection and the Patient Teaching Connection. (The actual skill of performing the skin puncture for the fingerstick blood sugar [FSBS] is taught in [Chapter 34, Skill 34.4.](#))

An abnormally elevated blood glucose level is known as **hyperglycemia**, and an abnormally low level is **hypoglycemia**. Both of these conditions can have serious effects on the patient, including falls, loss of consciousness, and even death. Hypoglycemia signs and symptoms may develop rapidly once blood glucose begins to drop. Signs and symptoms of hypoglycemia for you to monitor include the following:

- Clammy skin
- Shakiness or nervousness
- Nausea
- Increased irritability
- Confusion
- Dizziness
- Headache
- Weakness/fatigue
- Seizures and unconsciousness (at a critically low level of blood glucose)

When you suspect a patient's glucose level is dropping, perform an FSBS. If it is below 70%, provide a small glass of juice (4 ounces), gel tube (as directed by manufacturer instructions), three to four glucose tablets (as directed by manufacturer instructions), or 8 ounces of low-fat milk (as long as the patient is conscious). The juice or milk will help to raise the glucose level quickly. Recheck the glucose level within 10 to 15 minutes to ensure it comes up to a safe level. If the next meal is not imminent (i.e., within the next half-hour), administer a protein and complex carbohydrate snack to help maintain the glucose level until the next meal. A common snack used in this situation is peanut butter or cheese with a slice of whole-wheat bread. Continue to monitor the patient closely for hypoglycemia.

If the glucose level is low enough that the patient is at risk for or is having a seizure or is unconscious, the treatment is to administer dextrose 50% (D50W) IV. (Refer to your facility policy for specific glucose levels and treatment orders.)

Treatment for hyperglycemia is insulin administration, which will depend on the glucose level. If you suspect a patient's glucose level to be elevated, perform an FSBS to verify it. Signs and symptoms of hyperglycemia include the following:

- Increased thirst/dry mouth
- Hot, dry, flushed skin
- Headache
- Frequent urination
- Elevated blood glucose level
- Fruity-smelling breath
- Confusion
- Coma
- Death

• WORD • BUILDING •

hyperglycemia: hyper - excessive + glyc - sugar + emia - blood

hypoglycemia: hypo - deficient + glyc - sugar + emia - blood

glycemic: glyc - sugar + emic - blood

Laboratory and Diagnostic Connection

Diabetes

In order to effectively manage insulin-dependent diabetes, also known as *type 1 diabetes*, it is imperative to maintain *glycemic control*, which refers to the blood glucose level. This is done by thorough monitoring of blood glucose levels on a prescribed basis, such as once daily or before meals and at bedtime. Blood glucose should remain within a prescribed range established by the patient's health-care provider. The 2018 American Diabetes Association guidelines state that the target range of 80 to 130 mg/dL before meals is acceptable for existing diabetics. Glucose testing provides valuable immediate information for patients to adjust their food intake, medications, or activity level and for preventing hypoglycemic and hyperglycemic events. Glucose in the blood freely enters red blood cells and attaches to hemoglobin molecules, producing glycated or glycosylated hemoglobin. Health-care providers evaluate a patient's long-term glycemic control by measuring glycated hemoglobin, with a laboratory test known as **hemoglobin A_{1c}** (Hb A_{1c}). A red blood cell has an average life span of 120 days, so the Hb A_{1c} measures the amount of glucose present over a period of 2 to 3 months, giving a better overall picture of glycemic control. In other words, the Hb A_{1c} monitors the long-term glucose level of the patient over the previous 90 days. In nondiabetics, Hb A_{1c} usually is less than 6% of the hemoglobin. The goal of diabetes treatment is to maintain an Hb A_{1c} less than 7%. For diabetics without adequate control, it can range as high as 12% to 20%.

Patient Teaching Connection

Diabetes

The diagnosis of diabetes can be overwhelming. It is a disease that requires new medications, development of an awareness of blood glucose levels, and modification of various aspects of the patient's lifestyle and nutritional habits. Eating a healthier diet is among the most significant modifications the patient with diabetes must make. It is important that the patient with diabetes understand the disease, how diet aids in symptom management, and how to prevent short- and long-term complications. Many patients with newly diagnosed diabetes erroneously believe that their disease is related to sugar intake and therefore should be alleviated by the diet.

Before discharge, you will need to teach the patient about effective meal planning and nutritional intake. Using the following patient instructions will help you emphasize the most important points about healthy eating:

1. Eat a well-balanced diet, which includes carbohydrates, moderate protein, and fat in sparing amounts.
2. Know the differences between simple and complex carbohydrates:
 - *Simple carbohydrates*, which are found primarily in fruits, fruit juices, processed baked goods, and sugar, are quickly converted to glucose and can be used to rapidly raise the blood glucose. There is no need to totally eliminate them from the diet, but simple carbohydrates should be ingested in moderation, and carbohydrate intake should be consistent from day to day to maintain better balanced blood glucose.
 - *Complex carbohydrates*, which are found in whole grains, vegetables high in fiber, and legumes such as dried peas and beans, require the body to work harder to break them down to use for energy. This helps to maintain a more consistent blood glucose level. The majority of ingested carbohydrates should be complex carbohydrates rather than simple carbohydrates.
3. Eat three meals and an evening snack daily. Some health-care providers may prefer you to eat an additional snack in midmorning and midafternoon. Each of these should contain some protein.
4. Do not skip meals.
5. Increase your fiber intake.
6. Reduce your fat intake by baking, broiling, or grilling foods.
7. Lose weight if you are overweight. Studies show that even a loss of 5% to 10% of total body weight is significant to help lower blood glucose.
8. Wear a diabetic identification bracelet or necklace for emergencies.

KNOWLEDGE CONNECTION

What are your responsibilities regarding monitoring dietary intake? What signs and symptoms might indicate that a postoperative patient's diet has been advanced too rapidly? What are the signs and symptoms of hypoglycemia? What is the first step you should take when hypoglycemia is suspected? Explain the significance of Hb A_{1c}.

EATING DISORDERS

Eating disorders are evidenced by extreme disturbances in eating habits to such a point that the

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Eating disorders are evidenced by extreme disturbances in eating habits to such a point that the patient's health can be severely affected or even result in death. The most common of these disorders are anorexia nervosa, bulimia nervosa, and binge eating disorder, all of which involve the patient's relationship with food, eating, and body image. Women and girls are more at risk for developing an eating disorder than are males. While most eating disorders occur during adolescence or early adulthood, it is becoming more common to see them in middle-aged women also. Eating disorders may result from either physical or psychological causes and result in both physical and psychological effects. These patients have an impaired perception of self and do not see their eating or lack of eating as problematic.

Anorexia Nervosa

Anorexia nervosa is characterized by an excessive leanness or wasting of the body, known as *emaciation*. There is relentless self-starvation in an effort to reduce the body weight below normal. These patients do not see themselves

accurately and perceive that they are overweight even when they are at the point of physical starvation. Given the freedom to choose their meals, patients who are anorexic will continue to restrict their intake. They will exhibit obsessive thoughts related to their body shape and weight and will display an intense fear of obesity. The following are physical symptoms commonly seen in the severely underweight anorexic patient:

- Brittle nails
- Dull, dry, brittle hair
- Amenorrhea
- Severe constipation
- Lethargy or fatigue
- Below-normal vital signs
- Muscle weakness
- Muscle wasting
- Anemia

A possible complication is electrolyte imbalances, such as abnormally low levels of calcium, magnesium, zinc, and phosphorus. Another complication can include dehydration and metabolic acidosis or alkalosis from overuse of diuretics and laxatives. Patients with anorexia may struggle with the disease for a decade or longer, resulting in death if medical treatment is not sought before the damage is too severe to reverse.

Anorexia nervosa is especially difficult to treat. Patients with this disorder are most effectively managed by an interdisciplinary team with very specialized knowledge and skills.

Bulimia Nervosa

Bulimia nervosa is a disorder of eating binges accompanied by some type of behavior that will help to get rid of some of the calories that were ingested while bingeing. These behaviors can include self- or medication-induced vomiting (known as *purging*), excessive exercise, fasting, or overuse of laxatives. Some patients will also abuse diuretics to reduce water weight. Psychologically, the bulimic patient is also obsessed with body shape and weight; the binge-purge cycles are triggered by stressful events or may be influenced by peer pressure in adolescents. The acts of bingeing and purging are generally hidden from friends and family. These patients may hoard junk food and hide it for later bingeing. Physical symptoms that may be observed in a patient with bulimia nervosa include the following:

- Chronic inflammation or soreness of the throat caused by retching and exposure to gastric acid during vomiting
- Diarrhea from laxative abuse
- Increasing dental decay from exposure to gastric acid when vomiting
- Indigestion
- Regurgitation of gastric fluids into the esophagus, known as *gastric reflux*
- Dehydration

Possible complications include renal damage, fluid and electrolyte imbalances, and gastroesophageal reflux disease. Patients with bulimia also may suffer from anorexia, increasing the extent and severity of damage and the risk of death.

Patients with bulimia nervosa pose a significant challenge for the nurses working with them. They are most successfully managed by an interdisciplinary team with expert knowledge and skills in managing this disorder.

Binge Eating Disorder

A third type of eating disorder is characterized by episodes of eating large quantities of food at one time, with a loss of control while eating. The person with binge eating disorder may eat very quickly and be to a point of discomfort but continue to eat. This disorder is the most common eating disorder in the United States. According to the National Eating Disorders Association, it is seen in 3.5% of women, 2% of men, and 1.6% of adolescents. Feelings of shame, distress, and guilt accompany binges and are present after the binges. However, in binge eating disorder, the person

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accompany binges and are present after the binges. However, in binge eating disorder, the person does not purge. This disorder is often associated with being overweight or obese. It is considered a severe, life-threatening eating disorder, but it is treatable.

Patients with this disorder can benefit by treatment from the coordinated efforts of a team with expert knowledge and skills in managing binge eating.

KNOWLEDGE CONNECTION

Describe the common symptoms of the patient with anorexia nervosa and of the patient with bulimia nervosa.

FOOD-DRUG INTERACTIONS

Certain drugs and foods can interact and alter the effects of medications. Some drug and food combinations result in less serious interactions and should simply be monitored. However, other food-drug combinations significantly change the drug's effect and can result in a serious adverse reaction, making it imperative to avoid the offending food while administration of the drug is necessary. Some of the general types of food-drug interactions include the following:

- Some medications can affect nutrient intake by inhibiting appetite, causing GI discomfort, or causing some other reaction that inhibits food intake by making it unpleasant or painful. Other medications can stimulate appetite, increasing the quantity of food ingested.
- Certain medications can alter the absorption, metabolism, or excretion of nutrients. Alternatively, some nutrients can alter the absorption, metabolism, or excretion of medications.
- Other food-drug interactions can create medication toxicity.

• WORD • BUILDING •

gastroesophageal: gastro - stomach + esophageal - related to the esophagus

Prevention of these adverse events is possible with education. You have a primary responsibility to be aware of potential food-drug interactions and to educate your patients to be alert to these possibilities. There are numerous food-drug interactions, and as more medications are developed, the potential for new interactions increases. It would be impossible to memorize all of these adverse combinations, so you must faithfully research each new or unfamiliar drug in a drug book before you administer it. Many computer programs and smartphone apps can be used to look up medications that are new to you.

Drug Effects on Food Intake

The side effects of medication, such as nausea, vomiting, altered taste sensation, anorexia, decreased salivation, and sores or lesions in the mouth or the upper GI tract, can inhibit appetite or make food intake painful. Some medications can be very sedating or cause the patient to feel confused; this in turn will interfere with nutritional intake. These side effects may be monitored and managed when they occur on a short-term basis without causing significant weight loss. However, when the side effects are chronic, a plan of intervention must be developed to prevent weight loss and emaciation. For example, chemotherapy patients with severe nausea or vomiting probably would require the use of *antiemetics*, medications that help to prevent or reduce nausea and vomiting.

Many medications, such as corticosteroids, some antidepressants, and the atypical antipsychotics, can cause unintentional weight gain. For example, corticosteroids dramatically increase the appetite of some patients, resulting in increased consumption of food. These patients can gain large amounts of weight, often more than 20 pounds, in a short period. It is important to teach patients about these potential problems and to devise strategies to manage them before the weight is gained. For some patients, such as those with *cachexia* (malnutrition and wasting associated with cancer), weight gain is desirable. These patients may be prescribed medications to increase their appetite.

Drug Effects on Absorption of Nutrients

Dietary effects that reduce medication absorption typically occur through the influence of increased

Dietary effects that reduce medication absorption typically occur through the influence of increased gastric motility, conditions within the GI tract that are either too acidic or too alkaline, or direct interaction of food with a medication. Most medications are absorbed in the small intestine and may be absorbed more quickly if the stomach is empty during drug administration. It may be best that certain medications be given on an empty stomach to enhance absorption. However, other medications, such as ibuprofen or aspirin, should be taken with food to decrease gastric irritation. Some medications are best absorbed in an acidic medium, but others may be damaged by acidity. Those that are damaged by acidity often are available in an enteric-coated formulation that dissolves in the small intestine, thereby promoting better absorption. High-fiber foods and diets may decrease the absorption of certain medications.

Medications also can influence the absorption of nutrients. Drugs that damage the intestinal mucosa, such as NSAIDs or antineoplastics (cancer drugs), and certain antibiotics can cause malabsorption. The binding of the drugs to nutrients impairs absorption of the drug and/or the nutrients. For example, some antibiotics bind to the calcium in foods or supplements, which reduces the absorption of both the antibiotic and the calcium. Certain medications, such as histamine-2 (H₂) blockers or proton pump inhibitors, can reduce stomach acidity and impede the absorption of vitamin B₁₂, iron, and folate. Finally, medications can directly impede nutrient absorption by interfering with intestinal metabolism.

Drug Effects on Metabolism of Nutrients

Metabolism of food and medications occurs through the use of enzyme systems in the small intestine and liver. Some medications can inhibit or induce the activities of these enzyme systems, impairing the metabolism of nutrients. For example, the anticonvulsant phenytoin induces the liver system that metabolizes folate, vitamin D, and vitamin K. Patients taking this anticonvulsant will need additional supplements of these vitamins.

Many other medications can affect metabolism by mimicking the body's natural hormones, such as corticosteroid drugs that mimic the body-produced hormone cortisol. The human body, when exposed to these medications, acts as if it is experiencing an additional release of hormone. In the case of long-term treatment with corticosteroids, the patient may experience weight gain, rounding of the face shape called *moon facies*, hyperglycemia, bone loss, and muscle wasting.

Foods or their components can affect the enzyme systems as well and interfere with medication metabolism. Grapefruit juice, for example, has been found to inactivate or inhibit enzymes necessary for the metabolism of a number of medications. As a result, levels of the medication can be greatly increased, yielding a much stronger physiological effect. Another high-risk interaction occurs between many foods and herbs and the drug warfarin (Coumadin), an anticoagulant that is prescribed for patients with a history of blood clots. The patient taking warfarin should avoid eating foods containing moderate to high levels of vitamin K. Vitamin K can significantly weaken the anticoagulant effect of warfarin, increasing the risk of recurrent blood clots. In addition, the popular herbal supplements ginkgo, ginseng, St. John's wort, dong quai, and garlic, among others, contain compounds that enhance the effects of warfarin, which could possibly result in bleeding or hemorrhage. Patients who take warfarin should be educated to limit or avoid these foods and herbs. This education is your responsibility and that of the health-care provider and the pharmacist. [Table 24.2](#) presents a sampling of foods and drugs that should not be administered at the same time and foods that should be completely avoided while taking a particular drug.

• WORD • BUILDING •

antiemetic: anti - against + emetic - vomit

Table 24.2

Selected Food-Drug Interactions

Food or Drink	Drug
Large amounts of potassium-rich foods: bananas, oranges, leafy green vegetables	Potassium-sparing diuretics: spironolactone (Aldactone), amiloride (Midamor), triamterene (Dyrenium)
Foods high in vitamin K: broccoli, spinach,	Anticoagulant: warfarin (Coumadin)

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Foods high in vitamin K: broccoli, spinach, cauliflower, kale

Anticoagulant: warfarin (Coumadin)

Grapefruit, grapefruit juice, alcohol

Statin drug for high cholesterol: atorvastatin (Lipitor)

Orange juice, large amounts of licorice

Beta blocker for angina and hypertension: atenolol (Tenormin)

Alcohol

Beta blocker for angina, hypertension, and migraine headaches: propranolol hydrochloride (Inderal)

Dairy products

Quinolone anti-infectives: ciprofloxacin (Cipro), levofloxacin (Levaquin)

Tetracycline anti-infectives: doxycycline (Doryx)

Antifungals: fluconazole (Diflucan), ketoconazole (Nizoral), itraconazole (Sporanox)

Apple juice, orange juice, grapefruit juice

Antihistamine: fexofenadine (Allegra)

CLINICAL JUDGMENT IN ACTION

Scenario: Your patient has type 2 diabetes and is on the endocrine unit. When you assess the patient, she states, "I don't feel well. I'm dizzy and really tired." When you assess her skin, it is cold and clammy.

- What are the cues in this situation?
- What is your hypothesis?
- What immediate action, if any, will you take for this patient?
- How will you evaluate the effectiveness of your actions?

Drug Effects on Excretion of Nutrients

Excretion refers to the process of eliminating a medication or its metabolites or nutritional components from the body, a function usually performed by the kidneys. The length of time it takes to excrete a medication influences the extent of the medication's therapeutic effects. Slowed or inadequate excretion can raise the blood level of a medication and cause toxicity as successive scheduled doses are administered. Rapid or increased excretion can limit or greatly reduce the medication's therapeutic effect.

Certain medications can alter the kidney's excretion of nutrients such as minerals. Some diuretics, such as furosemide (Lasix), can cause increased excretion of important minerals such as potassium, sodium, chloride, calcium, and magnesium. Long-term use of furosemide requires regular monitoring of the patient's fluid and electrolyte status and usually requires potassium supplementation. Similarly, isoniazid (INH), a medication used to treat tuberculosis, can lead to increased excretion of vitamin B₆. Patients receiving INH therapy must be treated for more than 6 months, which can lead to vitamin B₆ deficiency. To prevent the deficiency, patients are started on B₆ as soon as INH is prescribed.

Many foods and nutrients can affect the excretion of medications. Lithium, a mood-stabilizing agent, is metabolized by the kidneys, where some of it is also reabsorbed and the remainder is excreted. This process is influenced by blood levels of sodium. Because lithium is similar in chemical structure to sodium, the kidneys often excrete lithium as if it were sodium. A person with a high level of sodium will excrete high levels of sodium and also of lithium, which can lead to subtherapeutic levels of the medication. Conversely, those with low sodium levels will retain extra lithium, leading to increased levels. Lithium is a medication that must be maintained in a specific therapeutic range, and increased levels can lead to adverse effects or even toxicity. This makes it extremely important to monitor the sodium levels in a patient taking lithium.

KNOWLEDGE CONNECTION

Which fruit juice is known to interfere with the metabolism of numerous medications? Patients taking warfarin should be educated to avoid foods with which vitamin? What side effects might you see in a patient taking long-term corticosteroids? Which vitamin must be supplemented in patients who are taking INH? Which electrolyte must be monitored in the patient taking lithium? Why is it important to educate patients about potential food-drug interactions?

• WORD • BUILDING •

subtherapeutic: sub – less than + therapeutic – treating

ENTERAL TUBES

Enteral tubes come in various styles and lengths and are used for multiple purposes. They may be used to decompress the stomach postoperatively, after abdominal injury, or as treatment for intestinal obstruction. These tubes may be inserted and used for nutritional support or medication administration. They may also be inserted to collect a specimen of stomach contents for diagnostic assessment of the GI tract, such as analyzing the gastric contents for pH or testing for blood.

KNOWLEDGE CONNECTION

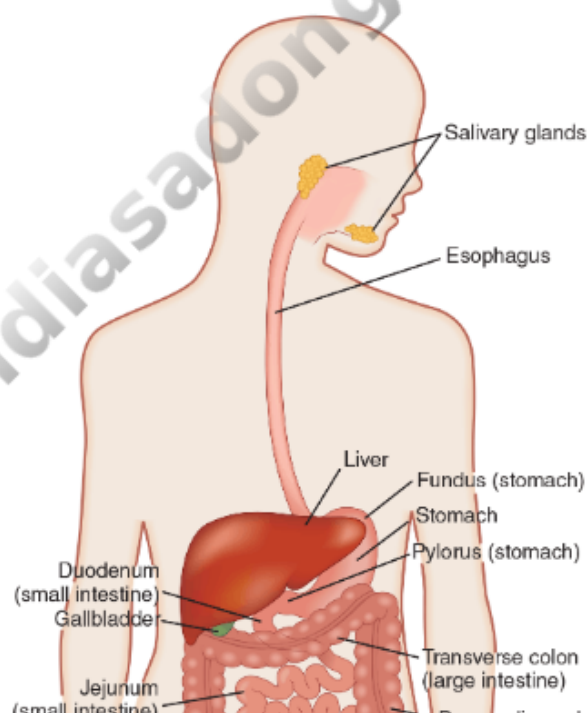
Explain four purposes for insertion of enteric tubes.

Anatomy and Physiology Connection

Gastrointestinal Tract

A review of the anatomy of the GI tract can help you to visualize the digestive organs and processes that occur as food moves through the body. The GI tract grows and changes through a person's life span, allowing greater digestion of food particles as it matures. This is part of the reason why solids are introduced to older infants, because the partially broken-down food particles may contribute to a heightened allergic response.

The organs that comprise the GI tract include the mouth; pharynx; esophagus; stomach; small intestine, which is further divided into the duodenum, jejunum, and ileum; and large intestine. The accessory organs include the liver and the gallbladder (Fig. 24.3). Digestion begins as food enters the mouth, where enzymes begin to break down the food, and continues as it travels through the GI tract. Along the way, peristalsis and the musculature of the GI tract guide the mechanical digestion; enzymes and secretions are added by various organs for chemical digestion. Nutrient absorption primarily occurs in the small intestine. Absorption is the process by which nutrients pass through the walls of the small intestine and enter into the bloodstream in the capillaries. Absorption occurs by varying mechanisms, which include passive diffusion and osmosis, facilitated diffusion, and energy-dependent active transport. The process of *passive diffusion* occurs when the pressure on one side of the membrane is greater and the nutrients move from the area of greater pressure through the mucosa to the area of less pressure. At this point, the nutrients enter the circulatory system and are used for energy or stored for later use. While most nutrients are absorbed in the small intestine, most of the water absorption occurs within the large intestine, helping to maintain water and electrolyte balance and to solidify the remaining waste product into soft, formed feces to be expelled via the rectum.



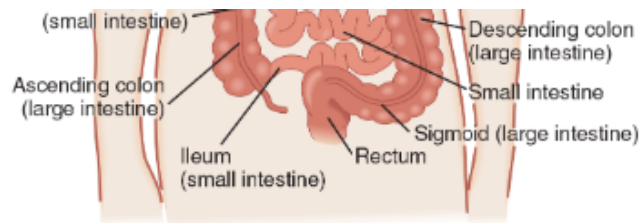


FIGURE 24.3 Organs of the digestive tract.

Nasogastric and Nasointestinal Tubes

The **nasogastric tube** (NG tube) is shorter than the nasointestinal (NI) tube and is inserted through the nose, down the esophagus, and into the stomach. The longer NI tube is also inserted through the nose and continues through the stomach, into the duodenum, or even farther into the jejunum. The terms *nasointestinal tube* and *nasoenteric tube* (NE tube) are often used interchangeably. The NI or NE tube usually has a smaller bore size and is more flexible than an NG tube. Some of them are weighted on the tip to help pull the tube through the stomach into the duodenum or jejunum; one commonly used NI tube is the Dobhoff ([Fig. 24.4](#)). It is mostly used for enteral feedings in patients who are at increased risk for aspiration, such as those with absent or diminished gag reflex or severe gastroesophageal reflux disease.

• WORD • BUILDING •

nasointestinal: naso - nose + intestinal - related to the intestine

nasoenteric: naso - nose + enteric - related to the small intestine

Tube Size and Length

The outside diameter of the NG or NI tube is measured using the French (Fr) scale: the larger the number, the larger the outside diameter of the tube. The most common sizes of feeding tubes measure 8 to 12 Fr with a length of 36 to 43 inches (90 to 108 cm). Larger-bore NG tubes, such as 16 or 18 Fr, are most commonly used for patients requiring gastric decompression or washing out of the stomach (termed *lavage*), obtaining gastric specimens, and instilling charcoal for the treatment of poisoning or drug overdose.

Single-Lumen Versus Double-Lumen

One style of NG tube is the Salem sump tube; it is a double-lumen tube, meaning that it has two separate channels inside the tube. One lumen is an air vent; the other vent is for drainage of stomach contents, irrigation, instillation of medications, and obtaining gastric samples. The vent lumen extends like a “pigtail” from the side of the tube near the end that connects to the suction tubing. Some manufacturers make the pigtail blue to differentiate it as the *air vent*.

Several small holes are distributed over 3 to 6 inches of the end of the NG tube lumen that is inserted into the stomach (refer back to [Fig. 24.4](#)). Once the suction begins to pull a vacuum through these small openings, it can cause the end of the tube to adhere to the wall of the stomach, serving to irritate the mucosal lining. If it remains adhered to the same spot for an extended period, it can damage the mucosa. The smaller second lumen usually is blue to differentiate it from the primary lumen. It is designed to allow for atmospheric air to be drawn into the tube to equalize the vacuum pressure in the stomach caused by the suction, helping to prevent the tube from adhering to the gastric mucosa.

A second type of large-bore NG tube is Levin’s tube, which has only one lumen. It can be used for the same purposes as the Salem sump tube, but it does not have the air vent lumen to allow equalization of vacuum pressure. Both the Salem sump and Levin’s tube are intended for short-term use only. Leaving them in place for an extended period can cause necrosis of the nasal passage. You will follow hospital protocol when replacing or rotating the Levin’s or Salem sump tubing.



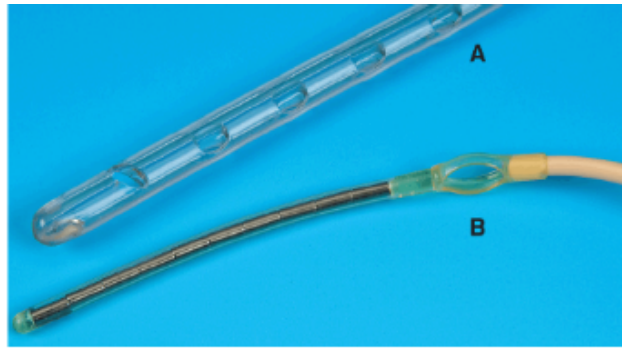


FIGURE 24.4 (a) Nasogastric tube: holes in the end of the tube allow contents to be suctioned from the stomach or fluid to be instilled into the stomach; (b) Weighted nasointestinal tube: smaller and more flexible than nasogastric tube, usually preferred for feeding (from Wilkinson JM, Treas LS, Barnett KL, et al. *Fundamentals of Nursing: Theory, Concepts & Applications*. Vol. 1. 4th ed. Philadelphia, PA: FA Davis; 2020).

GASTRIC DECOMPRESSION

Gastric decompression is the process of reducing the pressure within the stomach by emptying it of its contents, including ingested food and liquids, gastric juices, and gas. It is often ordered when the patient is experiencing persistent vomiting. This may be accompanied by abdominal distention, which may be caused by a partial or complete bowel obstruction. The appearance of the vomitus can often provide clues to the cause of persistent vomiting, so it is important for you to assess it and document your findings.

- Bright red blood indicates bleeding in the esophagus.
- Dark red blood indicates bleeding in the stomach.
- A coffee-ground appearance indicates blood that has been exposed to stomach acid; it has come from the esophagus and upper stomach and has reached the lower stomach and been mixed with stomach acid.
- Greenish-yellow liquid indicates bile. This usually means that the stomach is empty and the patient is vomiting the contents of the first part of the small intestine, the duodenum.

Insertion of an NG tube to low intermittent suction is part of the treatment for any of these conditions and for partial or complete bowel obstruction.

KNOWLEDGE CONNECTION

What does it mean to decompress the stomach? Describe the purpose of a double-lumen NG tube.

Insertion of NG Tube for Gastric Decompression

Placement of an NG tube requires a health-care provider's order, normally stating the type and size of tube and the purpose for the tube. If the health-care provider does not include this information, clarify all three pieces of information before inserting the tube. After reviewing the order on the medical record and clarifying it is necessary, review the record for conditions that might contraindicate insertion of the tube.

Preparing the Patient

Ask the patient whether they have ever experienced an NG tube insertion. Then explain the process to the patient, being honest but choosing your words carefully and observing the patient's responses. This helps you to know when the patient does not understand something, when they have a question, and when the patient begins to exhibit fear or anxiety. Your words and approach can have a dramatic impact on the patient's response. You can help a patient to better understand what you are about to do, gain the patient's confidence and cooperation, and alleviate some of their anxiety. However, if you fail to choose your words carefully, offer an incomplete explanation of what you are about to do, appear to be in a hurry, or display an apathetic or thoughtless attitude, you can raise their fear and anxiety level. This can result in a lack of patient cooperation that can make the insertion process much more uncomfortable for the patient and more difficult for you.

make the insertion process much more uncomfortable for the patient and more difficult for you.

Preparing Supplies for Use

Gather your supplies, wash your hands, verify the patient's identity, provide for privacy, raise the bed to working height, and lower the rails. (See [Skill 24.3](#) for a list of supplies and equipment that you will need.) Talk to the patient while you are preparing your work area and during the actual insertion of the tube. This will help to reduce the patient's anxiety and keep the patient focused on what you are saying rather than stressing as he or she watches you.

Arrange your supplies so that they are ready to use. Tear two pieces of 1-inch-wide silk tape, one piece approximately 1 inch long and the second piece 3 inches long. Beginning at one end of the 3-inch piece of tape, tear the tape lengthwise down the middle for approximately two-thirds of its length, leaving two 1/2-inch-wide tails that are about 2 inches long. The other end of the tape should still be 1 inch wide for about 1 inch of the tape's length. You will use the tape to secure the tube to the patient's nose; if your facility stocks them, you may obtain the adhesive securing device designed specifically for this purpose. Squeeze a small amount of water-soluble lubricant on a paper towel for lubricating the tip of the NG tube; place it within easy reach. Place your stethoscope on your neck so that it is ready for you to raise to your ears. Place a protective pad or towel over the patient's chest to prevent soiling their gown. Ask the patient to hold a finger against one side of their nose and exhale. Then have the patient repeat the action on the opposite side. Ask which side feels the most open. This will be the side you will want to use for inserting the NG tube.

Using the NG tube, measure the length from the tip of the patient's nose to the earlobe and then from the earlobe to the xiphoid process. Mark this spot on the tubing using either a red or black indelible marker. Mark the NG tube boldly at the point of insertion. This mark will be used as a point of reference for subsequent assessments of tube placement. You may also use a piece of tape, but do not wrap the tape around the tube so tightly that it is difficult to remove after the tube has been inserted.

Obtain an irrigation syringe that:

- Will hold 50 or 60 mL
- Has a plunger that is used to fill and empty the syringe
- Has a catheter tip or slip-tip so that the tip will fit into the NG tube opening

Connect the irrigation syringe to the open end of the NG tube and pull the plunger about halfway back to allow room for gastric contents. Then lay the tube with attached syringe within easy reach. Connecting the irrigation syringe before insertion of the tube will prevent gastric juices from soiling the bed, requiring an extra linen change. Provide the patient with an emesis basin and tissue, rinse-free bath wipe, or washcloth to hold in the nondominant hand and a glass of ice water with a straw to hold in the dominant hand. Apply your clean examination gloves.

Inserting an NG Tube

Establish with the patient some type of hand signal to use if the procedure becomes too uncomfortable or the patient would like you to pause. Assist the patient to a high Fowler's position. Tell the patient that you will tell them when to take a drink and swallow. Explain that if they can do this each time they are cued, it will make the tube go down much easier and faster.

Have the patient hyperextend the head just slightly. Gently insert the lubricated tip of the NG into the intended naris ([Fig. 24.5](#)). Remember that the nasal passageway begins to descend within a couple of centimeters. Avoid pushing the tip of the tube upward against the top side of the nasal passage, as it is quite painful and can damage tissue. Never force the tube. You should feel the tube begin to curve downward toward the nasopharynx. If not, gently lift the portion of the tube in your hand so that it directs the tip of the tube downward in the back of the patient's nasal passage. As soon as you reach the point of stimulating the patient's gag reflex, stop advancing the tube and pull it back just a tiny distance, about 1/8 inch (0.3 cm), *just to the point that the gag reflex is not being stimulated*, and instruct the patient to take a deep breath and relax. After giving the patient a couple seconds to relax, instruct the patient to begin drinking from the straw. Then, as the patient swallows, advance the tube, cueing the patient to "swallow ... swallow." The advancement of the tube coordinated with swallowing allows normal peristalsis to help pull the NG tube along. Repeat this process until the tube is inserted to the marked level. If the patient begins to gag or cough or

gives you the designated hand signal to stop, pause and allow the patient to relax and breathe. Coax the patient along in this manner until the insertion is completed. **Safety: If the patient gags or coughs continually and does not seem able to stop, the tube may be curled up in the back of the throat. Use a flashlight and tongue blade to view the posterior pharynx to look for the tube. If the patient cannot speak, becomes cyanotic, or coughs incessantly, the tube likely is in the lungs. Immediately remove the tube.** (Refer to [Box 24.3](#) for more on what to do when an NG tube enters the lungs.)



FIGURE 24.5 Gently insert the tip of the nasogastric tube into the most patent naris. As the tube begins to advance through the oropharynx, encourage the patient to sip water and swallow. This produces peristalsis, which helps to advance the tube into the stomach.

Box 24.3

Reacting When an NG Tube Enters the Lungs

When inserting an NG tube, it is important to know the signs and symptoms of the tube entering the lungs and the interventions to take if you believe this has occurred. Signs and symptoms of a tube entering the lungs include the following:

- The patient becomes cyanotic
- The patient is unable to make a verbal sound
- The patient is coughing spasmodically and uncontrollably

If any of these signs and symptoms are present during NG tube insertion, the tube has likely entered the lungs. If this occurs, immediately and completely withdraw the NG tube. Once removed, the patient will most likely resume normal breathing and regain their proper coloring. However, if the patient continues to show signs of distress, take appropriate steps to restore their airway and breathing.

SECURING THE NG TUBE. Once the tube is where you want it, remove the tape marker and apply the two pieces of tape you prepared to secure the tube to the patient's nose. There are various methods of taping the tube to the nose. Once you have managed several NG tubes, you probably will find that you prefer a particular method over the others. Here is one common way to apply the two pieces of tape. Apply the end of the 3-inch strip of silk tape (which you previously tore down the middle) to the patient's nose with the two tails hanging down toward the NG tube. Then wrap one of the tails around the tube, spiraling tape down the tube until the entire tail of tape is secured to the tube. Then wrap the second tail in the opposite direction around the tube in the same manner. Finally, apply the other piece of tape that is 1 inch wide and 1 inch long directly across the end of the first piece of tape on the patient's nose. This helps to further secure the tape to reduce risk of the NG tube coming out unintentionally. **Safety: Never let go of the tube until it is secured. It can come out in the blink of an eye, especially if the patient coughs, gags, or moves.** The patient would be very upset if you allowed the tube to fall out and have put the patient through a repeat insertion. **Safety: Be certain to tape the tube so that it is centered in the naris opening and does not press against any portion of the naris. If the tube presses against the nose, it can cause a pressure ulcer.**

pressure ulcer.

VERIFYING PLACEMENT OF THE NG TUBE. In some hospitals, it may be policy to check the pH of the aspirated NG tube contents to determine placement. The pH measurement should be less than 5.5. If you are unable to obtain gastric contents, turn the patient to the right side and recheck for gastric contents in 20 minutes. However, the most reliable way to check the placement of the tube is by x-ray. The tube has a radiopaque line that shows up on x-ray, confirming the location of the NG tube tip. **Safety: X-ray confirmation of NG tube location should always be performed immediately after the insertion of any type of enteral tube, whether it is placed for gastric decompression, gastric analysis, or administration of medications or feedings if unable to verify placement using the pH measurement.**

[Skill 24.3](#) provides step-by-step instructions regarding how to insert an NG tube.

Attaching the NG Tube to Suction

Attach the primary lumen to a connecting tube that extends from the suction device. Set the suction on low continuous or intermittent suction according to the health-care provider's orders. High suction is typically used only with intermittent suction.

Maintaining Patency of an NG Tube

To prevent an NG tube from clogging, it is important to assess the tube for *patency*, or the condition of being wide open, every 2 hours. Observe for the continual fluctuation and movement of gastric contents through the NG tube and connecting tube into the drainage container indicating patency. The health-care provider may write an "as needed" order for irrigation of the NG tube to maintain patency. [Skill 24.4](#) explains how to irrigate an NG tube.

Real-World Connection

Inserting an NG Tube

A patient presented to the emergency room with complaints of severe pain, nausea, and vomiting for more than 6 hours. After examining the patient and performing a couple of diagnostic procedures, it was determined the patient was suffering from an obstruction of the large intestine that was causing the portion of the bowel proximal to the obstruction to distend as it filled up with gastric juices and gas. Normal peristalsis was attempting to move the intestinal contents from the stomach through the small and large intestines but was unable to do so because of the obstruction, which explained why the patient was experiencing so much pain and nausea.

The patient was admitted, and orders were written to place the patient on NPO status, start IV fluids, and insert an NG tube connected to low intermittent suction. While the nurse was preparing to insert the NG tube, the patient vomited several hundred milliliters of brown semiliquid fecal material. Once the tube was successfully inserted, the extreme abdominal distention, pain, nausea, and vomiting began to improve immediately. With the administration of an analgesic and antiemetic, the patient got almost total relief by the time she had been there 1 hour. Within that first hour, a total of 2,000 mL of gastric contents and brown, foul-smelling, semiliquid stool were suctioned from the patient's GI tract. That particular patient later told the nurse that insertion of the NG tube was the best thing that had ever happened to her and she knew that it had saved her life.

Nursing Responsibilities for an NG Tube Connected to Suction

Your responsibilities when caring for an NG tube connected to suction include the following:

- Assess the tube every 2 to 4 hours to ensure patency.
- When necessary, irrigate a clogged tube according to the health-care provider's orders and facility policy.
- Monitor vacuum source settings for correct high or low suction and intermittent or continuous suction every 2 to 4 hours.
- Assess tubing connections to prevent accidental disconnection.
- Assess the color, amount, and consistency of gastric drainage.
- Assess the tube positioning and method of securing the tube so that the tube does not put pressure on the naris, which can result in a pressure ulcer.
- Manually clamp off the NG tube and auscultate bowel sounds every 4 hours, noting hyperactive, hypoactive, and absent sounds.
- Assess the abdomen for abdominal distention and palpate the abdomen noting firmness

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- Assess the abdomen for abdominal distention and palpate the abdomen, noting firmness.
- Assess the patient for reports of nausea, vomiting, pain, cramping, abdominal fullness, and discomfort every 2 hours.
- Assess for passage of rectal flatus, which may indicate the return of the bowel function known as *peristalsis*.
- Provide mouth care and apply lip moisturizer every 2 hours to prevent drying of the mucous membranes.
- Provide ice chips if allowed to keep the mouth moist.
- Monitor I&O to prevent fluid volume deficit or overload.
- **Safety:** *Because the suction will deplete the potassium from the stomach, monitor the serum potassium level, noting a level below normal, known as hypokalemia.*

Removing an NG Tube

Once the health-care provider has ordered the NG tube to be discontinued, it will be your job to remove it. This procedure is not complicated and takes only a few minutes to accomplish. Explain the process to the patient to alleviate any anxiety and gain the patient's cooperation. Provide the patient with a tissue to wipe mucus from the nose. Wear protective gloves and position a waterproof pad across the patient's chest to prevent soiling of the patient's gown. Instill 10 to 20 mL of air into the NG tube's main lumen to empty the tube of any gastric contents. This prevents fluid from dripping onto the epiglottis during removal. Remove the tape securing the tube to the patient's nose, pinch off the tube, and ask the patient to hold their breath to close off the epiglottis. Twist the tube gently to loosen any dried secretions that may be making the tube stick to the lining of the nasal passage or throat, and quickly withdraw the tube from the patient's nose. Wrap it up in the waterproof pad you positioned on the patient's chest and dispose of it in a biohazard bag. You will want to remove it from the patient's view because it may be covered with unsightly mucus and secretions. Assist the patient to wash the face and perform mouth care. Document the procedure and the patient's tolerance of it. [Skill 24.5](#) presents more information on removing an NG tube.

KNOWLEDGE CONNECTION

Why is it so important to adequately explain what you are about to do before inserting an NG tube? Why should you insert the tip of the irrigating syringe into the distal end of the NG tube before insertion of the NG tube? Why should you cue the patient to "swallow ... swallow ... swallow" sips of water when you begin to advance the tube past the oropharynx? Why is x-ray confirmation of tube placement so important? Why must you monitor the potassium level of a patient with an NG tube connected to suction?

ENTERAL NUTRITION

Patients you care for are often too weak or too ill to take in sufficient nutrients by eating. Certain illnesses can interfere with eating, swallowing, digestion, or absorption of food to the degree that the body is not receiving adequate nourishment. In these cases, nutritional support, in the form of liquid formulations of nutrients, can be administered to meet a person's energy needs. The two types of nutritional support are enteral nutrition and parenteral nutrition.

Enteral nutrition uses the GI tract as a delivery system and involves tube feedings that usually replace all oral intake, but it also may be given as a supplement to oral ingestion of nutrients. Enteral nutrition is the preferred method for patients with high nutritional needs and an intact GI system, such as those with burns, trauma, severe malnutrition, or neurological problems that impair swallowing. Choosing an enteral feeding route depends on a number of factors. These include the patient's medical condition and stability, current nutritional status, and nutrient requirements for healing and recovery. Feeding by the enteral route has been shown to improve nutritional status, strengthen immune response and reduce sepsis, minimize the hypermetabolic response to trauma, and maintain the function of the intestinal tract.

Feeding Tubes

As you read earlier, NG and NI tubes inserted for instillation of nutritional feedings are generally softer and smaller than those used for gastric decompression or lavage. They are inserted using basically the same method described previously; however, they tend to go down easier because they are smaller and softer.

they are smaller and softer.

• WORD • BUILDING •

percutaneous: per – through + cutaneous – skin

Percutaneous Endoscopic Gastrostomy and Jejunostomy Tubes

Patients requiring feedings have surgical endoscopic placement of a percutaneous feeding tube. *Percutaneous* means that an incision is made through the skin and abdominal wall, allowing the tube to extend from either the stomach or the jejunum to the external surface of the abdomen. If the tube is placed in the stomach, it is known as a **percutaneous endoscopic gastrostomy tube** (PEG tube) (Fig. 24.6). If the tube is inserted into the jejunum, it is called a **jejunostomy tube**, or J-tube. Figure 24.7 shows the insertion sites of different tubes. The tube is secured by tight suturing of the incision around the tube. This also helps to prevent leakage of formula during feedings. For patients with a newly inserted PEG tube or J-tube, the nurse must provide care for the surgical incision until it is completely healed. Of all the feeding tubes, the PEG tube is the most comfortable for the patient. Originally it was used for those patients needing long-term nutritional support. However, current trends in addressing patient comfort and improved technology have made this a good option for short-term therapy. The PEG tube or J-tube should also be marked at the point of insertion into the skin and measured from the point of insertion to the distal end of the tube. These two pieces of data should be recorded for use as a reference point when checking placement of the tube before instillations of feedings or medications.

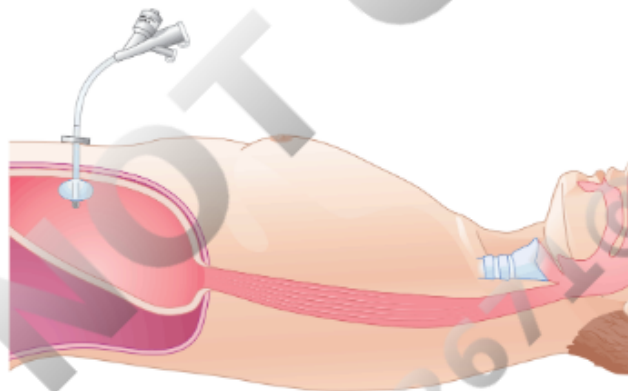
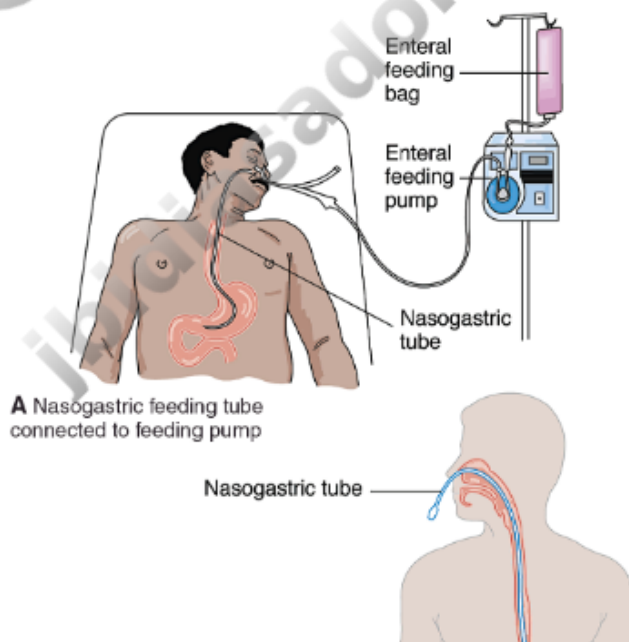


FIGURE 24.6 A percutaneous endoscopic gastrostomy tube (from Wilkinson JM, Treas LS, Barnett KL, et al. *Fundamentals of Nursing: Theory, Concepts & Applications*. Vol. 1. 4th ed. Philadelphia, PA: FA Davis; 2020).



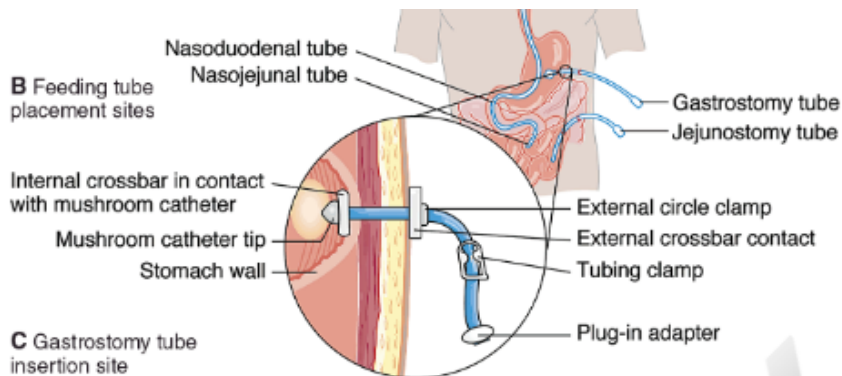


FIGURE 24.7 Feeding tubes: (a) Nasogastric tube connected to feeding pump; (b) Feeding tube placement sites: gastrostomy, jejunostomy; (c) View of gastrostomy tube insertion site showing the integument, subcutaneous, and muscle layers, in addition to the stomach wall, which a percutaneous gastrostomy tube must go through before entering the stomach (from Williams LS, Hopper PD. *Understanding Medical-Surgical Nursing*. 6th ed. Philadelphia, PA: FA Davis; 2019).

Some gastrostomies and jejunostomies may not have a tube inserted. Instead, they have a button placed at skin level that goes through the abdominal wall into the stomach or intestine and is held in place with a balloon. The tubing to deliver the enteral feeding attaches to the button (see [Fig. 24.8](#)).

KNOWLEDGE CONNECTION

Discuss the various types of feeding tubes. When is enteral nutrition the preferred feeding method?

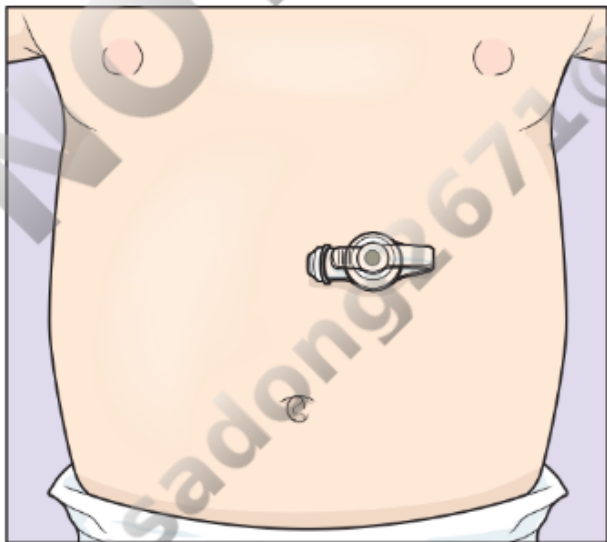


FIGURE 24.8 Gastrostomy button.

Tube Feeding Formulas

The selection of formula will be dependent on where the tube is placed and where formula will enter the GI tract. The patient's ability to digest and absorb nutrients will determine which type of formula is used. [Table 24.3](#) describes the most common types of enteral feeding formulas.

Table 24.3

Types of Enteral Feeding Formulas		
Type of Formula	Formula Description	Use for Patient Diagnoses
Standard formulas	Contain protein isolates from milk, soybeans, or a combination of purified proteins; provide 1 kcal/mL of solution and are proportionally balanced, delivering	Patients who can digest and absorb nutrients without difficulty but are unable to eat or

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	solution and are proportionally balanced, delivering 12%–20% of kcal as protein, 45%–60% of kcal as carbohydrates, and 30%–40% of kcal as fats; also contain vitamins and minerals and meet the nutritional needs of most patients	immediately but are unable to eat or swallow sufficiently Patients who need additional nutrition above their oral consumption
Hydrolyzed formulas	Contain macronutrients that are <i>predigested</i>, or have been partially or fully broken down to components that require little digestion before absorption; are often low in fat and may be enriched with medium-chain triglycerides to ease digestion and absorption	Patients who have compromised digestive or absorptive functions
High-protein formulas	Contain normal nutrients with additional protein	Burns Large open wounds Malnutrition
Diabetic formulas	Contain reduced levels of simple carbohydrates	Diabetes mellitus type 1 Diabetes mellitus type 2
Renal formulas	Contain reduced levels of potassium, sodium, and nitrogen	Renal insufficiency Renal failure
Pulmonary formulas	Provide 55% of the calories as fat so that less carbon dioxide is produced as oxygen is consumed	Lung cancer Chronic emphysema Chronic asthma Pulmonary fibrosis
Fiber-containing formulas	Contain higher levels of fiber	Diverticulosis Colon cancer Diabetes Heart disease Those in long-term care

Enteral Feeding Methods

The delivery method is chosen with consideration of a number of factors. These include the type and location of the feeding tube, the patient's medical condition and status of the digestive system, the type of formula being administered, and the length of time the patient has been on enteral feedings.

Feedings may be ordered for intermittent infusion and administered over 30 to 60 minutes or ordered continuously to be delivered over an 8-hour or even 24-hour period. The infusion rate will be regulated by either an infusion pump or gravity drip set according to the health-care provider's order.

Regulations by The Joint Commission require the use of new feeding tube, syringe, and feeding set (bag and tubing) connections, called ENFit connectors, for small-bore feeding tubes. This was put in place after studies and surveys showed that confusion existed between IV tubing and enteral feeding tubing, and when medications were drawn up in syringes that were compatible with IVs, the medications were sometimes accidentally given through the IV tubing instead of through the enteral feeding tubing. These tubes will only connect with each other and will not connect with IV tubing, Luer lock syringes, or slip-tip syringes. [Figure 24.9](#) shows an ENFit connection.



FIGURE 24.9 ENFit connector. (Borrowed with permission from the Global Enteral Device Supplier Association.)

Intermittent Tube Feedings

Intermittent feedings are used in patients who are on long-term enteral feedings and are no longer acutely ill. These patients may be in a long-term health-care facility, in a hospital, or at home. Intermittent feedings are administered in equal portions periodically throughout the day. The patient may receive a feeding of 300 to 400 mL of formula every 4 to 6 hours depending on the health-care provider's order. If the formula is delivered by gravity via a bolus or drip set or with a feeding pump, it usually is administered over a 30- to 60-minute period. The amount of formula delivered per feeding will depend on the total quantity of formula the patient needs to receive in a 24-hour period. Feedings are divided into equal portions and scheduled over the course of the day or scheduled only during waking hours to provide patients with uninterrupted sleep time. This latter schedule is preferable for those patients on long-term feedings, as this allows the patient a schedule that more closely resembles normal.

Intermittent feedings are generally well tolerated; maintaining the formula at room temperature decreases the potential for gastric discomfort. **Safety: Because of the high risk of aspiration with enteral feedings, the head of the bed should be elevated at least 30 to 45 degrees during the entire feeding and for 1 hour after the feeding to help reduce the risk of reflux and aspiration.**

BOLUS FEEDINGS. Bolus feedings are a type of intermittent tube feeding that is frequently used in which a health-care provider-ordered volume of formula is administered at set intervals throughout the day. To administer a bolus feeding, a large irrigating syringe (50 mL or greater) is attached to the feeding tube. The size of the feeding tube will determine the size and type of syringe tip that can be used: a catheter-tipped 50- or 60-mL irrigation or piston syringe for larger tubes or a 20- to 30-mL slip-tipped injection syringe for smaller tubes. If using the larger catheter-tipped irrigation syringe, remove the plunger from the syringe barrel and then insert the tip of the syringe into the open lumen of the feeding tube. Bend or crimp the tubing just below where it attaches to the syringe. This will prevent formula from flowing through the feeding tube before you are ready. Flush the tube by pouring 50 to 100 mL of sterile or tap water into the syringe depending on the situation. Release the bend or crimp in the tube so that water can flow through the feeding tube and then recrimp the tube. Carefully begin to pour the formula into the syringe barrel until it is almost full. Then release the bend or crimp in the tubing, allowing formula to flow through the feeding tube until the syringe is almost empty ([Fig. 24.10](#)). At this time, you may stop the flow by bending the

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until the syringe is almost empty ([Fig. 24.10](#)). At this time, you may stop the flow by bending the tubing again while you refill the syringe, or, if you are steady enough, you may continue pouring formula into the syringe as it is instilling until feeding is completed. Use whichever method works best for you, but *use caution to prevent air from entering the tube in between refilling the syringe*. Instilling air will increase gastric distress and should be avoided. This method should permit you to instill the feeding over a short period, generally 15 to 20 minutes, directly into the stomach or small intestine.

Feedings will be scheduled four to six times per day. **Safety: Residuals should be checked before administering each feeding (see the section “Checking Residual Gastric Volume” later).** Bolus feedings can be poorly tolerated, causing a number of problems such as cramping, gastric distention, diarrhea, and aspiration. For further information regarding administration of intermittent bolus enteral feedings, see [Skill 24.6](#).



FIGURE 24.10 Nurse administering intermittent bolus percutaneous endoscopic gastrostomy feeding (from Wilkinson JM, Treas LS, Barnett KL, et al. *Fundamentals of Nursing: Theory, Concepts & Applications*. Vol. 1. 4th ed. Philadelphia, PA: FA Davis; 2020).

Continuous Infusion Feedings

Continuous feedings are best administered via an infusion pump at a constant rate over an 8- to 24-hour period. The continuous infusion method promotes better tolerance of feedings and nutrient absorption. Use of the pump provides greater control of the rate and amount of feeding delivered. This method is used with critically ill patients because of the decrease in hypermetabolic response to stress, the risk for aspiration, and the incidence of diarrhea. Continuous feedings should be interrupted every 4 hours to check placement and residual volume, to administer water for hydration if ordered, and to flush the tube to prevent clogging. **Safety: These patients must have the head of the bed raised at least 30 degrees continually to reduce the risk of aspiration.**

Continuous tube feedings are typically initiated at full strength but at a slower rate to allow the intestinal tract time to adapt. You will monitor the patients for symptoms indicating poor tolerance, such as nausea, diarrhea, abdominal cramping, vomiting, or high gastric residuals. Once the patient exhibits tolerance of the feeding, the hourly rate can be increased every 8 to 12 hours until the desired rate of infusion is achieved. [Skill 24.7](#) provides information on how to use a feeding pump.

Checking Placement of a Feeding Tube

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You read earlier in the chapter that you must assess placement of any type of NG or NI tube before instillation of medication, irrigant, or feedings. It is so important that we want to emphasize it once more. **Safety: If the placement of a tube is not confirmed before administering a feeding or medication, there is a risk of instilling the formula or medication directly into the lungs via an NG tube or into the sterile peritoneal cavity via a PEG tube or J-tube.** Some patients have developed pneumonia—and other patients have actually died—from failure to confirm tube placement before instillations.

Checking Residual Gastric Volume

Before you can safely administer a tube feeding, not only must you check tube placement but you also must assess the *residual gastric volume*, or the amount of formula that still remains in the stomach from the previous feeding. If the previous feeding has not been absorbed and remains within the stomach and you proceed to instill another formula feeding, the additional formula can fill the stomach, which increases the risk of reflux via the cardiac sphincter into the esophagus. The esophagus lies in the immediate vicinity of the opening to the larynx at the superior end of the trachea, allowing refluxed formula to be aspirated into the lungs. Aspiration of fluids can initiate laryngeal spasms and/or result in pneumonia. **Safety: It is vital that you check the residual volume and tube placement before instilling each tube feeding.**

The process of checking residual volume is not complicated. Aspirate contents through the tube until the stomach is empty. If the residual amount is greater than the syringe can hold, measure the aspirate volume using a graduated container. Do not discard the residual aspirate, as it is rich in electrolytes and should be reinstalled into the stomach to prevent an electrolyte imbalance unless the health-care provider's orders state otherwise. Determine whether the residual volume exceeds the health-care provider's order for the amount at which to withhold the feeding. If the health-care provider does not write specific orders for handling residuals, then follow the facility's policy and procedure. However, if the health-care provider fails to provide specific instructions and the facility's policy and procedure manual does not specify a residual volume at which to withhold the feeding, then you must rely on what you learned as a student. Checking residual volume should also be documented ([Fig. 24.11](#)).

For intermittent feedings, check to see if the residual volume is 150 mL or less. If so, you should reinstall the aspirated contents and continue with the tube feeding. If the amount is greater than 250 mL, follow the directions for the patient receiving continuous tube feedings.

If the patient is receiving continuous tube feedings, determine the amount of residual volume and check the patient for any distention or abdominal pain. If the residual is less than the flow rate for 1 hour or less than 150 mL, proceed with the continuous feeding. If the residual is greater than 250 mL and there are no symptoms, reinstall the residual and check the patient again in 1 hour. If the residual is still greater than 250 mL, stop the infusion for 4 hours, assess the patient, and recheck residual volume. If the residual is still greater than 250 mL, call the health-care provider. If it is less than 250 mL, restart the infusion at 50% of the original rate. **Note: Always follow any health-care provider orders or facility policy for amounts of gastric residual volume. Use these guidelines only if no other guidelines exist.**

Pt Sel Logged in: Andrews, Lisa Documenting in current time and date **Log Off**

Patient Charting Vital Signs Order Entry Order Review Patient Teaching Patient CARD Patient Data M.A.R. Care Plans Reports

To Begin: Double-Click the Report

Admission History
Care Plans
Discharge Plan
Discharge Summary/
Instructions
MAR
Nurse's Notes
Patient Card
Patient Teaching
Vitals/I & O

Nurse's Notes

Nurse: Lisa Andrews Patient: Katell, Sam
Number: N2 Andrews Number: 08

Physical Assessment

Phase: 0

Nutrition and Hydration

Nutrition

NG Tube in use

mm/dd/yyyy 1600 NG tube placement confirmed with gastric pH of 4. Head of bed elevated to 30°. 10mL of residual gastric contents noted on aspiration. Fed 240mL of Ensure Plus plus 100mL of water flush after Ensure. Tolerated with no discomfort or flatulence. Head of bed remains at 30°.

FIGURE 24.11 Sample electronic health record documentation.

Nursing Care of the PEG Tube or J-Tube Insertion Site

Inspect the insertion site each day, noting any redness, tenderness, irritation, or drainage that might indicate infection. A new PEG tube or J-tube site should be cleaned daily with gauze and sterile saline. Using a circular motion, cleanse from the inner aspects of the site to the outer aspects. Ensure that the site is thoroughly dry and then cover it with a split-gauze dressing, also known as a *drain gauze*, which fits around the tube. For existing mature sites, cleansing should occur daily with a clean washcloth, warm water, and soap. The inner aspects of the site should be cleaned with a cotton swab to remove any drainage or leakage. Allow the site to dry thoroughly and leave it uncovered.

Nursing Responsibilities for a Patient With a PEG Tube or J-Tube

Your responsibilities for a PEG tube or J-tube include the following:

- Assessing tube placement by checking the tube markings to determine that the tube has not shifted or dislodged before each instillation
- Maintaining patency of the tube
- Performing daily assessment of skin integrity surrounding the tube
- Performing daily cleansing of the tube insertion site as appropriate
- Assessing bowel sounds daily, noting absent, hypoactive, or hyperactive sounds
- Assessing the abdomen, noting distention
- Assessing for gastric residual before instillation of medications or formula
- Elevating the head of the bed during feedings and for 30 to 60 minutes afterward
- Monitoring intake and output to ensure proper balance
- Monitoring weight, noting any loss
- Monitoring for diarrhea or constipation that might result from certain feeding formulas

COMPLICATIONS ASSOCIATED WITH TUBE FEEDINGS

Tube feedings are an artificial means of feeding a patient, and complications can arise. Common complications such as nausea and diarrhea are related to the patient's inability to tolerate the type of formula. The problem usually can be resolved by changing to a different formula.

Clogged feeding tubes are a problem that can stem from the processes associated with the feedings. It is important to prevent clogging by flushing the tube before and after instillations. If the feeding tube does clog, you can use a specialized plastic tool with spikes on either end, called a *declogger tool*, to help break up the clogged areas in the tube ([Fig. 24.12](#)). If the clog cannot be removed by the declogger tool, obtain a health-care provider's order to irrigate the tube to restore patency.

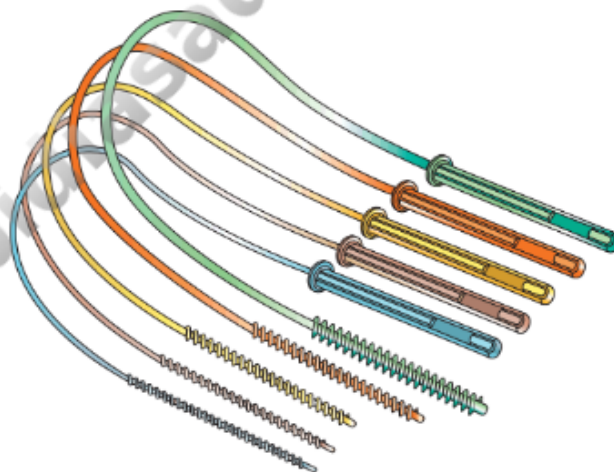


FIGURE 24.12 Declogger tool. Specialized plastic tool with spikes on either end that helps break up the clogged areas in feeding tubes.

Patients receiving tube feedings may have impaired swallowing ability because of stroke or debilitation from chronic illness, or they may be unconscious. *Aspiration* is an adverse event that places these patients at risk for additional serious illness. Aspirated material can include oropharyngeal bacteria, particulate matter, and acidified gastric contents. Aspiration of bacteria can cause aspiration pneumonia. Aspirated particulates can cause severe laryngospasm and bronchial constriction, wheezing, chest pain, and even respiratory failure. Aspiration of acidified gastric contents produces a chemical burn to the alveolar walls, causing rapid movement of protein and fluids into the alveoli. This results in a decline in lung compliance, with respiratory failure ensuing. The nurse can undertake a number of preventive measures to avoid aspiration ([Box 24.4](#)).

Metabolic problems are also potential complications: hyperglycemia, dehydration, fluid overload, or electrolyte imbalances can occur in patients receiving tube feedings. The proper choice of formula, a slow rate of infusion, and ensuring adequate hydration with water can help minimize and eliminate these complications.

Another serious complication associated with tube feedings is contamination of formula. Feeding formulas should be kept refrigerated until you are ready to use them. Contaminated formula can cause severe GI problems, including cramping, vomiting, and diarrhea, or food poisoning. Your patients receiving enteral feedings are already seriously medically compromised, and contaminated formula can lead to serious adverse consequences. Patients tolerate room-temperature formula better than cold formula, so you will need to plan to take formula out of the refrigerator and allow it to warm to room temperature approximately 1 hour before administering it. [Box 24.5](#) presents key points related to managing formula.

KNOWLEDGE CONNECTION

When is enteral nutrition the preferred feeding method? Why cannot all medications be administered via an enteral tube?

Box 24.4

Techniques to Prevent Aspiration

Because aspiration may be life-threatening, you are responsible for implementing interventions to reduce the risk of aspiration. This is a partial list of interventions that you should implement.

- Use the smallest diameter feeding tube possible.
- Confirm tube placement with x-ray immediately after insertion and before initiating tube feeding.
- Confirm tube placement before each instillation of medication, water, or formula.
- Make certain that active bowel sounds are present before instillations.
- Assess for abdominal distention before feedings.
- Elevate the head of the patient's bed to 30 to 45 degrees during intermittent feedings and at least 1 hour afterward; for continual feedings, the head of the bed must remain elevated at 30 to 45 degrees.
- Check gastric residual volumes before intermittent feedings and every 4 hours during continuous feedings.

Box 24.5

Tips for Managing Tube Feeding Formulas

Tube feeding formulas should be carefully managed to prevent spoilage, just as you protect the food you serve to patients and your own family. Patients receiving tube feedings may have compromised immunity, making it an even higher priority to prevent bacterial contamination of the formula. Here are some basic tips to remember.

- Wash hands thoroughly before handling formula or the feeding system.
- Rinse and wipe the top of the can before opening.
- Label all opened formula with the patient's name, date, and time when opened; follow the

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- Label all opened formula with the patient's name, date, and time when opened; follow the label instructions regarding the length of time you may use the formula once opened.
- Make sure all open cans are covered and stored in the refrigerator when not in use.
- Do not use unlabeled formula and out-of-date formula. Discard it in an appropriate receptacle.
- Rinse the feeding bag and tubing with clean water before adding new formula.
- Flush the tube with water before and after each use.
- Hang only the amount of formula that can be administered in 6 hours; discard formula that hangs for more than 6 hours.
- Never add new formula to old.
- Change the feeding bag and syringe every 24 hours.

Supervision/Delegation Connection

Tube Feedings

Assistive personnel work under the supervision of the registered nurse (RN) or licensed practical/vocational nurse (LPN/LVN). Tasks that may be delegated include administration of enteral tube feedings via an NG tube, NI tube, PEG tube, or J-tube. The delegation of feeding can occur only after verification of the feeding tube placement. You are responsible for this assessment and cannot delegate performance of tube placement verification. You also are responsible for supervising the assistive personnel in preparing and positioning the patient to receive the tube feeding. The assistive personnel can document the intake and output and report these data. Remember that the rules governing delegation are dictated by each state's Nurse Practice Act. It is your responsibility to know the role and scope of practice for both yourself and the assistive personnel you supervise and to delegate nursing care appropriately.

PARENTERAL NUTRITION

Parenteral nutrition is administered directly into the bloodstream via a central venous catheter (CVC), bypassing the GI tract. Parenteral nutrition can supply complete nutrition, including amino acids, dextrose, emulsified fats, vitamins, minerals, and trace elements, to the patient who cannot meet their nutritional needs via the oral route. **Safety: Aseptic technique is required, and the bag of solution must be changed every 24 hours.** It is used for patients with one or more problems that are complicated by malnutrition and low protein levels. These patients generally do not have a functioning GI tract or are unable to absorb adequate nutrients through enteral feedings. [Box 24.6](#) presents a partial list of conditions in which parenteral nutrition might be appropriate.

The administration of parenteral nutrition depends on the patient's nutritional needs and the expected length of time parenteral feedings will be required. **Partial parenteral nutrition (PPN)** may be administered through a peripherally inserted central catheter inserted into a smaller peripheral vein. **Total parenteral nutrition (TPN)** may be administered through a CVC placed in a larger central vein. The core differences between TPN and PPN are the vessel selected to receive the nutrition and the concentration of solution infused. Only the "least hypertonic" solutions may be infused into the peripheral veins, whereas it is possible to deliver higher concentrations through the larger central veins.

Box 24.6

Conditions That Might Require Parenteral Feeding

There are many medical conditions for which parenteral feeding may be required. Here is a partial list of conditions that may be treated with parenteral feeding:

- Burns or trauma, severe
- Cancer
- Chemotherapy
- Hepatic failure
- Intestinal obstruction
- Intractable vomiting or diarrhea
- Malabsorption disorders
- Necrotizing enterocolitis

- Paralytic ileus, unrelenting
- Renal failure
- Short bowel syndrome

Partial Parenteral Nutrition

PPN is used for those patients who are able to meet some of their nutritional needs orally but require additional calories or nutrients for a limited period because of their illness. PPN does not meet the patient's total nutritional needs without concurrent ingestion of food via the GI tract. Patients with short bowel syndrome or malabsorption syndromes often require PPN in addition to oral intake. Solutions infused into peripheral veins must be isotonic or only mildly hypertonic to prevent phlebitis and decrease the risk of thrombus. Parenteral nutrition solutions include a mixture of dextrose, amino acids, lipid emulsions, electrolytes, vitamins, and trace minerals in sterile water. The Infusion Nurses Society states that peripherally administered PPN solution cannot contain a concentration that exceeds 10% dextrose and 5% amino acids. Because of these restrictions, PPN will provide a limited amount of calories, necessitating oral supplementation for the patient to have adequate daily caloric intake.

Total Parenteral Nutrition

TPN is a hypertonic, nutritionally complete solution that is delivered via a large-diameter central vein. It provides calories, essential vitamins, minerals, and electrolytes and restores nitrogen balance, promoting tissue growth and wound healing, which makes it useful in patients who are nutritionally depleted. Infusion into a large vein allows for the TPN solution to be diluted quickly. TPN is used when a patient's nutritional requirements are high and it is anticipated that the need will be long term. Patients for whom it may be used are those being treated for extensive trauma, burns, or sepsis; those undergoing aggressive cancer treatment; or those with GI illnesses that interfere with metabolism and absorption of nutrients.

The CVC that is used for TPN can have two, three, or four separate lumens, or channels ([Fig. 24.13](#)). It must be inserted surgically into one of the larger central veins, such as the subclavian vein or internal jugular vein, and advanced until the catheter tip is positioned in the superior vena cava ([Fig. 24.14](#)). The insertion site where the catheter perforates the cutaneous layer must be continually covered with a sterile dressing, usually a transparent occlusive dressing. Often the patient who is a candidate for insertion of a central line for TPN already has a suppressed or impaired immune system. Because this extremely invasive procedure places the patient at risk for infection, you should receive specific training before providing nursing care for a central line. Some facilities limit care of central lines to RNs only, while others permit LPN/LVNs once they have been trained. It is your responsibility to know the facility policies and procedures where you are employed, in addition to knowing what is allowed by the Nurse Practice Act in your state.

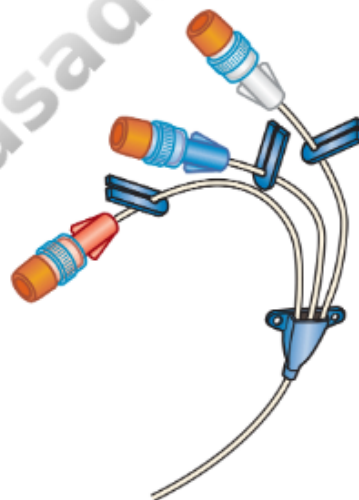


FIGURE 24.13 Triple-lumen central venous catheter injection ports (from Wilkinson JM, Treas LS, Barnett KL, et al. *Fundamentals of Nursing: Theory, Concepts & Applications*. Vol. 1, 4th ed. Philadelphia, PA: EA Davis; 2020).

Vol. 1. 4th ed. Philadelphia, PA: FA Davis; 2020).

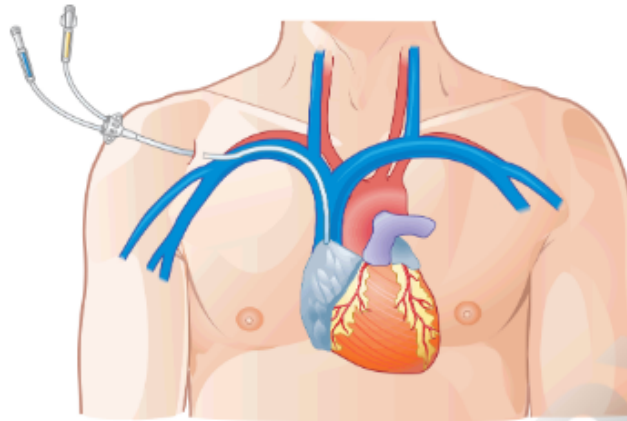


FIGURE 24.14 Central venous catheters are typically inserted into larger central veins such as the subclavian or jugular veins and threaded into the superior vena cava (from Wilkinson JM, Treas LS, Barnett KL, et al. *Fundamentals of Nursing: Theory, Concepts & Applications*. Vol. 1. 4th ed. Philadelphia, PA: FA Davis; 2020).

Patients receiving TPN may receive their medications mixed into their solution or intravenously infused into the designated lumen or a port linked to the central line. They often receive insulin to help control serum glucose because the TPN solution has a high glucose concentration. In general, because of the potential for incompatibilities, the addition of medications to TPN solution should be limited, and compatibility must be ensured. TPN is expensive, administration requires that the patient be continually monitored while receiving TPN, and TPN increases the risk for the development of infection or mechanical or metabolic problems. Therefore TPN is discontinued as soon as the patient's nutritional needs have been met or the patient can tolerate enteral or oral feedings.

TPN Infusion

Initially, TPN is infused slowly to give the body time to adapt to the infusion of a highly concentrated glucose solution. It is increased in increments of 25 mL per hour until the desired rate of infusion is achieved. It is administered via a continuous drip infusion on a pump to ensure a controlled rate of infusion. It is essential that infusion be slow and controlled because a rapid infusion of TPN solution can cause severe hyperglycemia or hypoglycemia, with the potential for further complications such as coma or death. Initially, a patient will receive 1 liter of solution infused over 24 hours, with the rate being increased by 1 liter per day until the desired volume delivery is reached.

When caring for a patient receiving parenteral nutrition, it is important that you monitor laboratory values closely to assess for fluid or electrolyte imbalance until the patient is fully stable on a parenteral formula. Before you can discontinue a patient's TPN, the patient must be gradually weaned from TPN to either enteral or oral feedings in order to prevent metabolic complications.

KNOWLEDGE CONNECTION

What is the difference between TPN and PPN? Why is TPN administered via a central line catheter? What laboratory values should be monitored in patients receiving TPN?

Monitoring Nutritional Status

Patients receiving parenteral nutrition should be monitored closely for nutritional status. This includes monitoring daily weights until stable and certain laboratory results. Electrolyte levels should be monitored daily for 1 week before decreasing to twice a week. Prealbumin, albumin, and total protein should be monitored, but they do not have to be monitored as frequently as the electrolyte levels. The glucose level should be monitored at least twice daily, usually four times, in order to determine how often the patient requires insulin, and how much, to maintain a stable blood glucose level.

Prealbumin is produced primarily by cells in the liver, but it also is produced by GI mucosa and the pancreas. It is quite sensitive to changes in nutritional status, has a relatively short half-life, and is

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pancreas. It is quite sensitive to changes in nutritional status, has a relatively short half-life, and is rapidly responsive to protein depletion. These characteristics make prealbumin a good indicator of acute changes, while albumin remains the standard for monitoring chronic malnutrition. Normal prealbumin levels are 12 to 42 mg/dL. FSBS should be assessed every hour during the initial TPN infusion and then every 4 to 6 hours thereafter to provide a reference for insulin administration.

KNOWLEDGE CONNECTION

Why do you need to monitor prealbumin in patients receiving parenteral nutrition? Why is it important to perform FSBS every hour initially and then every 4 to 6 hours afterward?

Settings Connection: Home Health

Enteral and Parenteral Therapy in the Home

Patients receiving enteral or parenteral therapy may be on these feedings for long periods. Many patients will not require lengthy hospital stays and will be discharged home on feedings. The home health nurse's role in management of the patient requiring nutritional support is one of patient/family advocate, coordinator, and educator. A key factor in the success of enteral or parenteral nutrition will be family or caregiver involvement. This should begin early in the process, ideally as soon as parenteral nutrition is implemented in the hospital setting. Caregivers should be involved in discussions regarding site selection, nutrient delivery method, and formula selection, with consideration given to their ability to manage the care and the affordability of the feeding formula. The nurse will educate the patient and caregiver regarding care of the feeding tube, tube placement site, flushing the tube, administering feedings, safety issues, and managing complications that may arise, such as a clogged tube. Safety in the administration of enteral and parenteral feedings is important. Caregivers must be taught how to verify the tube placement, how to minimize risk for aspiration, and how to manage air in the PPN or TPN line. The nurse will also teach the caregiver how to safely store formula and minimize the risk of contamination. Signs and symptoms of problems for which the patient should be monitored must be taught and made available in printed format for later referral.

The home health nurse will follow the patient to assist with management of feedings as the patient transitions to home. However, the nurse will gradually assume a more supportive role as they teach the caregiver to assume responsibility for the feedings and care and management of the tube. Once the caregiver has assumed full responsibility, the nurse will remain involved and continue to follow the patient, assessing for adequate nutrition as evidenced by maintenance of weight and normal laboratory values, including glucose and albumin levels. The nurse will also perform physical assessments regularly, monitoring skin integrity by looking for areas of pressure ulceration or risk for breakdown, checking for signs of muscle wasting, monitoring the lungs for signs indicating aspiration of particulates and evolving pneumonia, and monitoring for signs of difficulty tolerating tube feedings, such as increased abdominal discomfort, bloating, or diarrhea. The nurse will serve as the primary communicator with the health-care provider and assist to coordinate care as the patient's condition changes.

Nursing Care Plan for a Patient Post-CVA With a PEG Tube

You have been assigned to provide care for 56-year-old Mr. Ayala, who has now had a PEG tube placed. He was admitted earlier in the week and is 5 days post-CVA, more commonly called a "stroke." He had an NG feeding tube placed first, but now has a new PEG tube. What do you need to know before you begin planning his nursing care? What questions might you want to ask yourself as you collect data? What are the areas in which the patient might have problems or be at high risk for developing problems? For Mr. Ayala, you would probably seek to answer the following questions:

- What is the status of his neurological and cardiovascular systems?
- What residual deficits from the stroke does he have?
- How is he coping with those deficits?
- Does he have impaired mobility?
- Is he at risk for injury?
- Is he at risk for aspiration?
- Is he continent of bowel and bladder?
- Is he able to communicate?

Assessment

Now that you have an idea of what data you need to collect, your initial step will be to review the

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Now that you have an idea of what data you need to collect, your initial step will be to review the current medical record. Specifically, you need to review the patient's medical history, noting medical conditions that alter the patient's nutritional needs and influence the type of feeding formula selected. While reviewing the patient's chart, observe for data that indicate possible complications or that may influence the timing of administration or formula selection, such as the following:

- Height and weight
- Aspiration risk
- Diarrhea or constipation
- Fluid or electrolyte imbalance
- Nausea and vomiting
- Skin breakdown

Your next step will be to review the patient's laboratory results for the following tests:

- Hemoglobin and hematocrit levels
- Blood glucose levels
- Protein level
- Fluid and electrolyte status

Review of the patient's vital signs shows that they are stable: blood pressure – 136/82 mm Hg; temperature – 99°F; pulse – 70; respirations – 16. His serum albumin is 34 g/L, and his electrolyte panel indicates normal results. Parts of his blood count, the hematocrit and hemoglobin, are within normal range, indicating no anemia, but there was no order to check his protein level. A continuous enteral feeding was initiated today, infusing at 20 mL per hour via a feeding pump. You noted in the chart that he is unable to speak, known as *dysphasia*, but he is able to answer questions by nodding or shaking his head.

Next you will need to review the patient's medication record, checking listed medications for incompatibility with feeding formulas; for example, those that may in some way affect GI tract functions, such as increasing or decreasing peristalsis. When you enter Mr. Ayala's room he is awake and alert and makes eye contact as you introduce yourself. He also attempts to smile; however, only the left side of his face moves. He reaches out with his right hand to shake your hand. He can use the urinal but requires assistance to get up to the bedside commode. You know that he has difficulty swallowing since the stroke, and a PEG tube was surgically placed yesterday. He shakes his head when you inquire if he is experiencing any pain or discomfort. He writes that he would like to be bathed and get up in a chair as soon as someone can help him. You explain that part of your nursing care will be to check his PEG tube and site to ensure that healing has begun, assess the security of the PEG tube, and check for signs of infection. You check the site and note an approximately 1/4-inch ring of mild erythema but no signs of purulent drainage or leakage of gastric contents. The tube is secured to the patient's abdomen with tape, and the feeding system is connected, infusing formula at 20 mL per hour. He shakes his head to indicate that he has had no nausea, vomiting, or diarrhea. You auscultate active bowel sounds in all four quadrants.

You previously noted health-care provider orders to advance the feeding to 30 mL per hour by 3 p.m. You end the assessment with an evaluation of his skin, noting a slight reddening of intact skin over the coccyx. The rest of the initial shift assessment findings are within normal limits.

Before leaving the room, you say good-bye to Mr. Ayala. You also lower the bed to its lowest position, raise the head of the bed to 30 degrees, confirm that its wheels are locked and side rails are up, and position his call light and tissues where he can easily reach them.

The new PEG insertion site, located on the left upper quadrant of his abdomen, exhibits no signs of infection or bleeding. His PEG tube is intact and patent, with a continuous infusion of formula at 20 mL per hour that needs to be increased to 30 mL per hour over the next 6 hours. He is continent of both bowel and bladder. He has a 2-cm reddened area on the coccyx that blanches and returns to normal color after repositioning. The data tell you that your patient is status post-CVA with stable cardiovascular status, vital signs, and glucose level. Thus far, he is tolerating his enteral feedings, he has had adequate I&O, and his fluid and electrolyte panels indicate adequate hydration and nutritional status.

Nursing Diagnoses

After analyzing the data, you determine that Mr. Ayala has a few high-risk and problem areas. Your

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After analyzing the data, you determine that Mr. Ayala has a few high-risk and problem areas. Your first priority is his safety, as he has right-sided hemiparesis and has difficulty swallowing. The appropriate nursing diagnoses for these problems include the following:

- Impaired swallowing
- Impaired physical mobility

The patient has two actual problems that place him at risk to develop additional problems. When there is evidence that a problem could develop, that is known as a *potential* problem. For example, the impaired swallowing places the patient in danger of aspirating, resulting in the additional nursing diagnosis “Risk for aspiration.” Because the patient actually suffers impaired physical mobility, this problem places him at risk for developing additional problems. Also, because he already has a new incision, he has impaired skin integrity.

Assessment of the coccyx found the skin reddened, further data reinforcing his risk for impaired skin integrity in the form of a pressure ulcer. This potential problem requires that action be taken to prevent its occurrence. He already has one site of impaired skin integrity—the new incision made to insert his PEG tube.

Planning

Your nursing care plan will include your role and that of the assistive personnel, CNAs, and other health-care providers working with you. Your plan of care will be tailored to meet Mr. Ayala’s needs; therefore the interventions will be specific to him. Your first priority will be his safety, then his nutrition, and next his activities of daily living. You must remain aware of his psychosocial needs, as he is experiencing role change, isolation, and impaired verbal communication. You need to provide care that will help him adapt to his body changes while preventing problems that are directly related to mobility impairments.

Because the top priority is impaired swallowing, your plan must ensure that the patient is identified as high risk for aspiration.

The patient also has impaired mobility, and because of this he is at risk for falls and for constipation.

The patient also has impaired skin integrity related to the newly inserted PEG tube and is at risk for infection at the site.

Implementation

Now that you have developed a plan, you need to put it into action. See later for the nursing interventions needed to help prevent possible problems and deal with his existing problems.

Evaluation

Now that you have enacted the patient’s care plan, you will want to evaluate its effectiveness and your patient’s progress toward health. You want to see if the interventions you have chosen and implemented have helped prevent further skin breakdown, promoted healing of his PEG site, and prevented him from aspirating. His problems are improving or resolving, and your plan of care has helped him improve.

Nursing Diagnosis: Impaired swallowing

Expected Outcomes: Will maintain patent airway. No aspiration.

Interventions

Evaluation

Note in the care plan that patient is NPO with only occasional ice chips. Post NPO sign on the door, above the bed, and in EHR.

Day 1: Eats ice chips sparingly. Remains NPO.
Day 3: No change.

Keep the head of the bed elevated a minimum of 30 degrees at all times.

Day 1: HOB maintained at 30 degrees. No signs of aspiration.
Day 4: No aspiration.

Reinforce NPO status, elevation of bed, and risk of providing oral fluids with the patient and family members.

Day 1: Verbally instructed patient, wife, grown daughter regarding importance of NPO status and risk of aspiration. Family verbally acknowledged understanding. Patient nodded agreement.
Day 2: Following instructions.

Assist the patient with mouth care to prevent dryness four times per day. Apply lip moisturizer to lips prn.

Day 2: Mucous membranes pink, moist, and intact. Able to apply lip balm unassisted.
Day 5: Mucous membranes remain pink, intact, and moist.

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Keep suction equipment at the bedside ready for use at all times. Discontinued Day 7	Day 1: Suction not needed. Day 2: No choking or aspiration. Day 3: Working to relearn swallowing technique with therapists. Day 6: Makes effort to swallow melted ice chips without problems. No suction needed. Day 7: Suction removed from room.
Keep the call light within reach at all times.	Day 1: Uses call light when assistance is needed. Day 5: Calls staff as needed.
Infuse Jevity per PEG tube with pump at 30 mL/hr. If tolerated for 24 hours without GI distress, increase to 50 mL/hr. If tolerated for 24 hours, increase to 75 mL/hr.	Day 1: Tolerated 30 mL/hr without nausea or distention; bowel sounds active in all quads except RLQ, which was hyperactive. Day 2: Tolerated 50 mL/hr. No distention, nausea, or diarrhea. Day 3: Rate increased to 75 mL/hr. Day 4: Active bowel sounds throughout; no reports of gastric distress.
Check PEG tube placement and residual volume every 4 hours. Follow health-care provider orders, hospital policy, or protocol in chapter for gastric residual volume.	Day 1: Placement confirmed; residuals 30 mL and less. Day 2: Placement confirmed; residuals 40 mL, 50 mL, 35 mL. Day 4: Residuals less than 50 mL consistently. Day 6: Residuals and placement checks unchanged.
Monitor I/O closely for 2 weeks; report variances.	Day 1: In - 880 mL; Out - 630 mL. Day 3: In - 1,925 mL; Out - 1,475 mL. Day 7: I/O balanced.
Record daily weights at 0800; report losses. Discontinued 08/10	Day 1: Admission: 179 lbs. Day 2: 177 lbs. Day 4: 177 lbs. Day 7: 177.5 lbs.
Nursing Diagnosis: Impaired physical mobility Expected Outcomes: During hospitalization: (1) Skin integrity will remain intact. (2) No development of contractures. (3) Will remain injury-free. (4) No development of constipation. Maintain bowel elimination with a BM at least every 3 days.	
Interventions	Evaluation
Assist the patient to get up into a chair a minimum of three times per day and to a bedside commode as needed.	Day 1: Requires 2 assists to stand and pivot to chair; reported fatigue afterward. Day 4: Assists with right arm; can bear weight on unaffected leg with 2 assists to balance. Day 5: Still unable to move left leg voluntarily; no longer reports fatigue after up. Day 7: Able to transfer to bedside commode with retractable arms, using short slide board.
Cue the patient to reposition every 2 hours; teach him the importance of activity. Discontinued Day 7	Day 1: Cued every 2 hr. Requires 2 assists to turn to either side. Day 3: Continues to require 2 assists to turn to right side but can turn to left side unassisted; needs help to position support pillows. Day 5: Continues to help reposition self and do a little more each day. Day 7: Moves about unassisted in bed; no longer requires staff to cue or assist.
Encourage and assist the patient as needed to perform active/passive ROM to all extremities twice a day.	Day 1: Active ROM exercises performed; 5 sets of each movement. Day 3: Active-passive ROM today; able to move unaffected joints and use right hand to move left arm and hand; states he feels so much better after exercises. Day 5: Reported sleeping better at night; believes it is due to ROM; working hard to move left leg; some left shoulder lift performed independently.

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	shoulder lift performed independently. Day 7: Increasing strength; slight movement of left arm.
Assess pressure points for erythema and apply lotion to back and all pressure points three times per day. Day 9: Modified to twice per day, during bath and p.m. care.	Day 1: Skin over pressure points intact; mild redness over coccyx, 2-inch circle, resolves 15 min after repositioning. Day 4: No redness noted on pressure points except pale pink area on right elbow where patient leans on elbow to sit up; gel-filled elbow pads applied. Day 6: No further redness of elbows or other pressure points. Day 7: Able to sit up without leaning on elbow; removed elbow pads; skin intact; no redness.
Keep a hand roll positioned in the left hand.	Day 1: States the hand roll keeps the left hand from sweating so badly. Day 3: Left fingers still flaccid; hand roll replaced daily. Day 7: No signs of contractures in left hand. Day 8: Exercises his left hand 8 to 10 times/day, even while watching TV and visiting; only places hand roll during sleep hours.
Keep the bed in its lowest position with the wheels locked.	Day 1: Bed kept in low position and brakes locked; does not attempt to get up unassisted. Day 5: Calls for assistance when he needs to get up.
Raise both bed rails continually.	Day 1: 2 rails up; doesn't try to get out of bed alone. Day 4: Pulls on bed rail with right hand to help turn him.
Assess bowel sounds with each tube placement check, noting hyper- or hypoactivity.	Day 1: Active bowel sounds in RLQ, RUQ, LUQ, and hyperactive in LLQ. Day 5: Active times 4 quadrants. Day 8: Hypoactive.
Monitor and document bowel movements for frequency, consistency, color, and amount.	Day 1: Chart shows last BM 2 days prior. Day 2: Continent of small, brown, soft BM. Day 3: None. Day 4: Large, soft dark brown BM. Day 5: 3 semisoft BMs. Day 7: None in 2 days. Day 8: Moderate amount soft brown BM.
Administer a laxative of choice, if there is no BM after 2 days.	Day 5: Having BMs without difficulty. Day 7: Laxative requested and given.

Skill 24.1 Assisting a Patient With Feeding

Assessment Steps

1. Review the patient's medical record **to determine diet orders**. Review the chart for problems that might impair the patient's ability to self-feed.
2. Assess the patient for risk of aspiration. **Patients at high risk are those with dysphagia, decreased gag or cough reflex, or decreased level of consciousness and those with difficulty managing saliva. If the patient is at high risk, ensure that suction is available at the bedside.**
3. Assess the patient's intake and output, weights, and laboratory values. **This helps to evaluate nutritional status.**

Planning Steps

1. Gather needed equipment and supplies: a protective barrier to protect the patient's gown and assistive devices. **Being prepared saves you time and effort and increases efficiency.**
2. Provide the patient an opportunity to use the bathroom. **This helps to reduce mealtime interruptions.**
3. Clean the overbed table of clutter and unpleasanties such as a urinal, toilet tissue, or emesis basin containing sputum. **A clean and neat table is more conducive to eating.**

Implementation Steps

1. Follow the Initial Implementation Steps located on the inside back cover.
2. Assist the patient to a chair or elevate the head of bed to the high Fowler's position. **This positions the patient for eating and decreases the risk of aspiration.**
3. Provide a warm, wet washcloth (one with soap and one without soap for rinsing) or a rinse-free bath wipe to wash the face and hands **to reduce microorganisms.**
4. Place a protective barrier beneath the patient's chin so that it covers the patient's gown **to prevent the gown from becoming soiled.**
5. Provide as much assistance as the patient needs while encouraging optimal independent feeding. **It is important to encourage optimal independence for the patient's self-esteem but to still ensure that adequate assistance is provided to optimize nutritional intake.**
6. Position the overbed table directly in front of the patient so that food is in front of the patient, as it is at one's own dining room table. **This helps to make it more like mealtimes at home and is more conducive to eating.**
7. If the patient is able to self-feed, assist with tray setup as needed: open containers, cut up meat, butter bread, and open condiments and apply them if the patient desires. If the patient is unable to self-feed, proceed with feeding the patient. **It is important to do everything possible to encourage nutritional intake.**
8. If possible, sit down while feeding the patient. **This shows the patient that you are not in a hurry, which encourages the patient to eat more.**

If the patient is blind:

- First identify whether the sight impairment is long-standing or a new loss. If the patient is not used to being blind, you probably will need to feed the patient or assist the patient in feeding. If the patient has been blind for a long time, you will need to ask the patient if you can help.
- Be certain to approach the subject judiciously and compassionately because the patient may be very sensitive about the inability to do certain things and the loss of independence.

If a blind patient is able to self-feed:

- Open containers and cellophane-wrapped flatware or condiments, cut meat, butter bread, and season food if the patient desires.
- Verbally inform the patient where each food item is located on the plate using the numbers on the face of a clock and the relationship of plates, bowls, glasses, and flatware for identifiers. For example: "Your meatloaf is at 12 o'clock, mashed potatoes are at 3 o'clock, green beans are at 6 o'clock, a glass of iced tea is on the right above your fork and spoon, a bowl of peaches is on the left above your knife, and apple cobbler is to the far left of the bowl of peaches. Here is your napkin."
- It is helpful to gently guide the patient's hands to feel the sides of the dinner plate as you begin to tell where the food contained on the plate is located. Also guide the patient's hand to feel the glass, flatware, and extra bowls. **These things will encourage more intake.**

9. Allow the patient to determine the order in which they will eat foods **to allow the patient more control.**

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10. Encourage the patient to handle easy-to-eat foods such as bread or small pieces of fruit **to promote independence.**
11. Keep food bits small and allow the patient adequate time to chew and swallow food **to reduce the risk of choking.**
12. Make mealtime a pleasant social occasion, engaging in conversation while feeding and encouraging the patient to talk about their life, children, or favorite hobbies. Smile during the meal. **This makes mealtime more conducive to eating.**
13. Follow the Ending Implementation Steps located on the inside back cover.

Evaluation Steps

1. Observe the level of independence and amount of assistance required **to determine whether additional assistance is needed.**
2. Observe for signs of difficulty swallowing, such as gagging, choking, or significant coughing, during oral intake **to prevent aspiration.**
3. Measure the amount of oral fluid intake each meal. Estimate the percentage of the solid foods eaten. Measure liquids in milliliters. Document this in the appropriate location in the chart.
4. Observe the patient's ability to tolerate their diet, monitoring for signs such as abdominal distention and complaints of cramping, nausea or vomiting, or indigestion, and document in the chart. **This helps to determine whether or not the diet should be changed.**

Sample Documentation

05/25/28 1230 Required total assistance with soft diet. Consumed 50% of potatoes, green beans, & chicken without swallowing problems. No complaints of GI distress. _____ Nurse's signature and credentials

Skill 24.2 Recording Intake and Output

Assessment Steps

1. Review the patient's medical record for previous intake and output (I&O) to see if there are imbalances. **This helps you to know if you need to encourage fluids or not.**
2. Introduce yourself to the patient and explain the procedure. Review the rationale for the monitoring of I&O so that the patient and family understand the importance of measuring. **This results in better compliance.**

Planning Steps

1. Gather needed equipment and supplies: paper and pen, specimen pan, urinal, graduated receptacle, and drinking glass with increments marked in milliliters. **Being prepared saves you time and effort and increases efficiency.**
2. Instruct the assistive personnel who will be assisting in the patient's care. **It is your responsibility to communicate the needs of the patient to assistive personnel.**
3. Place paper and pen at the bedside for recording I&O **to increase compliance.**

Implementation Steps

1. Follow the Initial Implementation Steps located on the inside back cover.
2. Measure, do not estimate, the volume of liquid consumed, except when the exact volume of an empty container is known, such as a milk carton that holds 240 mL. If only part of the liquid was consumed, measure what is left and record the difference **to increase accuracy.**
3. Document on the I&O sheet in the room; at the end of the shift, transcribe the total to the flow sheet. **Documenting at the bedside increases accuracy.**
4. If intake volumes are less than desired, encourage the patient to drink more. Find out the patient's likes and dislikes and provide the preferred items as long as they are not contraindicated by their diet order. If necessary, offer a few ounces every hour to get adequate fluid in the patient. **Don't wait until the end of the shift to become concerned about inadequate intake. Be proactive. This helps to more accurately determine fluid balance.**
5. Measure the amount of urine and other sources of output, such as the amount emptied from drainage devices, stomach aspirate, emesis, or diarrhea excreted by the patient, and document on the flow sheet in the chart. **This helps to more accurately determine fluid balance.**
6. Document all output in the room immediately after measurement. Avoid trying to remember totals until the end of the shift. At the end of the shift, transcribe the totals to the flow sheet in the chart **to increase accuracy.**

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increase accuracy.

7. Follow the Ending Implementation Steps located on the inside back cover.

Evaluation Step

1. Compare the I&O amounts, noting any significant imbalance. **The output total should be within 300 to 500 mL of the intake, allowing the difference for the insensible losses.**

Sample Documentation

At the bedside:

Date Jan 1

Shift 3/11

Output		
Intake	Urine	Emesis
240 mL	255 mL	130 mL
70 mL	155 mL	
240 mL	530 mL	
85 mL	940 mL	
90 mL		
725 mL		

Then, the end-of-shift totals are transcribed to the I&O sheet in the chart as follows:

Date	Shift	Oral	IV	Irrig	Total Shift Intake	Urine	Emesis	NG	Drains	Total Shift Output
10/1	7a-7p	1,050 mL	240 mL		1,290 mL	1,115 mL	275 mL			1,390 mL
	7p-7a	725 mL	240 mL		965 mL	940 mL	130 mL			1,070 mL
	24-hour Total									

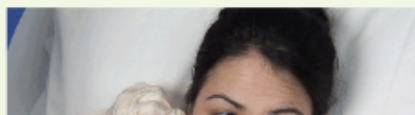
Skill 24.3 Inserting a Nasogastric Tube

Assessment Step

1. Review the medical record for medical history, current diagnoses and problems, and conditions that might limit the patient's ability to cooperate during the procedure or that contraindicate insertion of an NG tube.

Planning Steps

1. Gather needed equipment and supplies: correct type and size of ENFit NG tube if for feeding, ENFit irrigation syringe, container of water, emesis basin, washcloth or rinse-free bath wipe, towel or waterproof pad, examination gloves, lubricating jelly, tape or device to secure tubing to nose, glass of ice water and straw, stethoscope, flashlight, tongue blade, suction setup with a long connector tube or ENfit feeding setup, indelible marker, and tape measure. **Being prepared saves time and effort and increases efficiency.**
2. Explain the procedure to the patient, discussing it fully and instructing the patient in ways in which they can assist to make the procedure go smoothly. **This reduces anxiety and increases patient compliance during the procedure.**
3. Place the patient in a high Fowler's position.
4. **If the NG tube is to be connected to suction:** Set up the suction apparatus, including a canister and connector tube that connects to the NG tube adaptor.
5. Measure the length of the NG tube by measuring from the tip of the patient's nose to the earlobe and then to the xiphoid process. Mark this length on the tube with a small piece of tape or an indelible ink marker. Do not wrap the tape so tightly around the tube that it is difficult to remove after the tube has been inserted. **This determines the approximate length to insert the NG tube.**

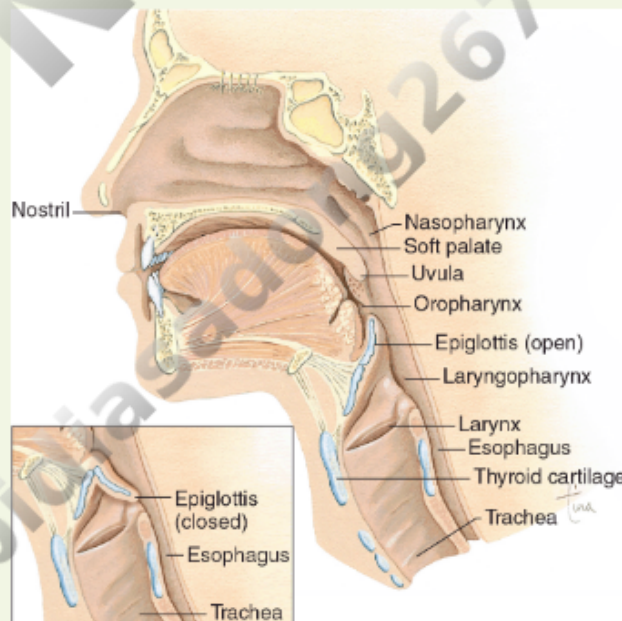




- Determine the patency of the patient's nares before selecting an acceptable naris. Select the nostril with the greatest airflow. Have the patient blow their nose.

Implementation Steps

- Follow the Initial Implementation Steps located on the inside back cover.
- Wearing clean examination gloves **to protect yourself from potential contact with body fluids**, pull the plunger back on the irrigation syringe **to make room for gastric contents to drain into the syringe**. Snugly insert the tip of the irrigation syringe into the open end of the NG tube. **This will prevent soiling the bed with gastric contents as the NG tube reaches the stomach, which saves you from having to change linens.**
- Lubricate 3 to 4 inches of the tip of the NG tube with a small amount of water-soluble lubricant. **This reduces friction and increases ease of passage of the tube. Safety: Avoid using petroleum-based lubricants because they increase the risk of pneumonia as a result of inhaling petrolatum products.**
- Repeat the instructions for the patient and ask if they have questions. **Because this procedure frightens many patients, repeating the instructions is sometimes helpful to alleviate anxiety.**
- Have the patient hyperextend the neck. Insert the tube through the patient's naris, slowly advancing the tube downward into the nasopharynx. You may feel slight resistance as the tube enters the nasopharynx; gentle rotation or changing the angle of the tube typically overcomes this. **Safety: Never force the tube if there is more than the slightest resistance. Forcing the tube can result in serious damage to the nasal mucosa. If you meet resistance, remove the tube, relubricate the end of the tube, and attempt insertion in the opposite naris.** It is typical for the patient's eyes to water during tube insertion. **Remember to be gentle and reassure the patient.**



- When the patient begins to gag, you know that the tube is in the posterior nasopharynx; stop advancing the tube and withdraw it a very short distance (about 1/8 inch), **just enough to stop the gagging**. Encourage the patient to take a couple of deep breaths and relax. Then instruct the patient to flex their head forward slightly, drink a sip of water, and swallow. **This position aids passage of the tube through the posterior pharynx and esophagus rather than the larynx.**
 - Continue to cue the patient to "swallow ... swallow ... swallow." Continue to gently advance the tube each time the patient swallows. After two to three advancements, allow the patient brief rest periods

each time the patient swallows. After two to three advancements, allow the patient brief rest periods (if needed) before resuming cues and advancement of the NG tube. If the patient continues to gag and cough, use a penlight and tongue blade **to make certain the tube has not curled up in the back of the patient's throat. Safety: If the patient exhibits symptoms of respiratory distress or cyanosis, stop the procedure and withdraw the tube immediately.**



- Monitor the depth of tube insertion, noting when you reach the marked level. Observe for return of stomach contents into tube and attached syringe.
- Once the tube is in place, secure the end by taping it to the patient's nose. As you tape the tube, make certain to avoid positioning the tube where it is pressing against the naris **to prevent accidental dislodgment of the tube.**



7. Apply a tubing clamp or insert a plug. **Safety: Until placement has been verified by x-ray, nothing can be instilled into the NG tube, including a normal saline flush, water, medications, and formula.**
8. Record this measurement in the electronic health record (EHR) **for comparison at future placement checks.** If verification cannot be made by pH placement, notify radiology and transport the patient to the radiology department for an x-ray. After tube placement has been verified, return the patient to their room and connect the NG tube to suction or feeding as ordered.
9. Follow the Ending Implementation Steps located on the inside back cover.
10. Document the procedure, time it was performed, patient's tolerance of the procedure, length of tube extending from the insertion site, method used to determine correct placement and findings, characteristics of any drainage, and the patient's current respiratory status.

Evaluation Steps

1. Confirm tube placement by x-ray or pH measurement **to ensure proper placement. The pH measurement should be less than 5.5.**
2. Assess the patient's level of comfort **to enable you to provide optimal comfort.**
3. Assess the characteristics of any aspirate or drainage: color, consistency, and amount.

Sample Documentation

05/31/28 1015 Explained procedure. States she has had an NG tube previously and understands the process.

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Inserted 18 Fr Salem sump NG tube into Rt. naris without resistance or difficulty, per health-care provider's order. 17.5-in NG tube extending from insertion site. Obtained 450 mL thick, dark green gastric contents within a few minutes. Denies discomfort. _____ Nurse's signature and credentials

05/31/28 1035 Tube placement confirmed in the stomach and verified pH measurement of 4.5 is within range . NG tube attached to low intermittent suction per health-care provider's order. _____ Nurse's signature and credentials

Skill 24.4 Irrigating a Nasogastric Tube

Assessment Steps

1. Check the patient's medical record for orders and a radiographic report verifying correct placement of the NG tube. ***This helps to prevent instilling irrigant into a displaced tube.***
2. Observe the NG tube and assess for secure placement ***to prevent accidental dislodgment of the tube.***

Planning Step

1. Gather needed equipment and supplies: catheter-tipped irrigating syringe or a 50-mL syringe with a smaller slip-tip if required to fit the NG tube to suction or ENFit catheter for a feeding tube, normal saline (NS), pH paper and color chart, tape measure, emesis basin or bag, towel or waterproof pad, examination gloves, lubricating jelly, and tape or a device to secure tubing to the nose. ***Being prepared saves time and effort and increases efficiency.***

Implementation Steps

1. Follow the Initial Implementation Steps located on the inside back cover.
2. Place the patient in a sitting or high Fowler's position ***to help reduce reflux and aspiration.***
3. Position a waterproof pad under the end of the NG tube and unclamp the end of the NG tube if clamped. Otherwise, disconnect the tube from suction or the feeding setup. ***This protects linens from soiling and allows access to the tube.***
4. To verify placement, assess whether the insertion mark is still positioned at the point of entry and measure the length of tube extending from the insertion site, comparing it with the recorded length. If your hospital uses the pH method for checking placement, use a syringe to withdraw 5 to 10 mL of gastric residual volume, observe the color of the contents, and measure the pH.
Patients who have been fasting at least 4 hours have a pH less than 5.5; those on continuous feedings will have a slightly higher pH. ***This ensures that the irrigant enters the stomach.***
5. Reinstill aspirated gastric contents ***to prevent electrolyte imbalance*** and disconnect the syringe from the NG tube.
6. Draw up the ordered volume of normal saline (usually 30 mL) into the syringe and reconnect the syringe to the NG tube.
7. Gently instill the normal saline into the tube. ***Excessive force increases risk of reflux and aspiration. Normal saline helps clear debris that may clog the tube and prevent it from draining effectively.***
8. Lower the end of the NG tube and gently aspirate the irrigant back into the syringe. ***Excessive suction can damage the gastric mucosa.*** Repeat steps 7 and 8 as needed until the normal saline flows freely into and out of the tube.

For a Double-Lumen (Salem Sump) Tube:

Draw up 30 mL of air into the syringe and inject it into the blue "pigtail." Always follow facility policy. ***This helps clear the air vent of secretions that could cause it not to function properly.***

9. Remove the syringe from the tube; clamp or plug the proximal end of the tube or reconnect it to suction. Secure the tube to the patient's gown.
10. Follow the Ending Implementation Steps located on the inside back cover.

Evaluation Steps

1. Assess the patient's tolerance of the procedure.
2. If the NG tube was connected to a suction source, observe for fluid movement through the tube ***to verify patency.***

Sample Documentation

06/04/28 0920 NG tube irrigated with 30 mL NS according to health-care provider's orders. Irrigated without

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06/04/28 0920 NG tube irrigated with 30 mL NS according to health-care provider's orders. Irrigated without difficulty. NG tube patent. Reconnected to low intermittent suction. No C/O discomfort during irrigation.

Nurse's signature and credentials

Skill 24.5 Removing a Nasogastric Tube

Assessment Step

1. Review the patient's intake and output *to determine fluid imbalances.*

Planning Step

1. Gather needed equipment and supplies: towel or protective pad; washcloth or rinse-free bath wipe; tissues; examination gloves; irrigation syringe; and the patient's toothbrush, toothpaste, and mouthwash. **Being prepared saves time and effort and increases efficiency.**

Implementation Steps

1. Follow the Initial Implementation Steps located on the inside back cover.
2. Place the patient in the high Fowler's position and hand the patient some tissues.
3. Clear the tube of any secretions by using an irrigation syringe to inject 10 mL of air through the main lumen. **This reduces the risk of fluid dripping into the airway as the tube is withdrawn.**
4. Remove all tape from the patient's nose and the NG tube **so that the tube can be removed.**
5. Have the patient hold their breath and then gently but quickly withdraw the tube, immediately wrapping it in the disposable protective pad to remove it from the patient's line of vision. **This reduces the risk of aspiration during removal of the tube.**
6. Discard all contaminated supplies in the appropriate biohazard waste receptacle.
7. Provide the patient with a warm, wet washcloth (one with soap and one without soap for rinsing) or a rinse-free bath wipe to wash the face.
8. Assist the patient as needed to brush the teeth and rinse the mouth with mouthwash **to rid the mouth of debris, mucus, and bad taste from the tube removal.**
9. Offer a glass of cold water to drink, if the patient is not NPO, **to soothe a dry throat.**
10. Follow the Ending Implementation Steps located on the inside back cover.

Evaluation Steps

1. Determine whether the patient is experiencing any discomfort or pain related to the procedure and medicate accordingly. **This provides for patient comfort.**
2. Assess for active bowel sounds and repeat in 4 hours while observing for abdominal distention, nausea, or vomiting. **This helps to determine GI tract tolerance of tube removal.**

Sample Documentation

11/21/28 0930 NG tube removed without difficulty per Dr's orders. Abdomen flat and soft; no complaints of nausea. Bowel sounds active in all 4 quadrants. Instructed to eat no more than 1 tsp of ice every 30 min and to report any nausea or bloating to staff. _____ Nurse's signature and credentials

Skill 24.6 Administering Intermittent Bolus Enteral Feedings

Assessment Steps

1. Review the patient's medical record and health-care provider's order to determine the rate, route, and type of formula to be administered and **to reduce the risk of error.**
2. Evaluate the patient's current nutritional status. Review the laboratory data and other available diagnostic data, such as fingerstick blood sugar readings. Assess for fluid volume excess or deficit and metabolic or electrolyte imbalances. **This helps to determine the patient's nutritional needs.**
3. Assess whether the patient has impaired swallowing or has had facial trauma or surgeries of the alimentary canal, head, or neck. **These conditions indicate the patient's need for enteral tube feedings.**
4. Obtain a baseline weight to determine increases or decreases after initiation of feedings. **This provides for later comparisons to help evaluate the effectiveness of feedings.**
5. Assess for active bowel sounds; if absent, withhold feeding until you check with the health-care provider.

provider.

Planning Step

1. Gather needed equipment and supplies: clean gloves; pH paper strips; prescribed enteral formula; an irrigation syringe or a 30-mL or larger syringe with a smaller slip-tip catheter tip if required for a snug fit into the NG, NI, or PEG tube for a large-bore feeding tube or an ENFit irrigation syringe for a small-bore feeding tube; a tape measure; and a cup of water. **Being prepared saves time and effort and increases efficiency.**

Implementation Steps

1. Follow the Initial Implementation Steps located on the inside back cover.
2. Raise the head of the bed 30 to 45 degrees **to reduce the risk of reflux and aspiration.**
3. Have the tube feeding formula at room temperature **to reduce intestinal cramping.**
4. Check the expiration date of the formula and the integrity of its container **to determine the safety of the formula.**
5. Arrange supplies on the overbed table within easy reach.
6. Check for correct tube placement and gastric content residual volume by assessing whether the insertion mark is still located at the insertion point, measuring the length of the tube from the point of insertion to the distal end, comparing it with the recorded measurement, and using the irrigation syringe to aspirate all stomach contents. **Safety: If the tube is a Salem sump tube, do not insert the syringe tip into the blue air vent.** Measure the amount of residual aspirate and check the pH with pH paper if your facility uses pH to determine placement. Reinstill aspirate **unless** the amount is 250 mL or greater, in which case you will check for abdominal distention. If there are no such symptoms, you will:
 - Reinstill the aspirate and wait an hour. Then repeat assessment of residual gastric volume. If it is still 250 mL or greater, call the doctor. If it is less than 250 mL, restart the infusion at 50% of the original rate or follow facility policy. **This helps to prevent electrolyte imbalance and prevent overfeeding the patient.**With a J-Tube: Proceed as noted earlier; however, you will aspirate intestinal secretions. Intestinal residuals tend to be small in amount and more alkaline. Residuals of more than 10 mL with an acidic pH may indicate that the tube has dislodged and is displaced to the stomach.
7. Shake the feeding formula to ensure that it is mixed well.
8. Remove the plunger from the irrigation syringe barrel and attach the tip to the feeding tube for a large-bore feeding tube. For a small-bore feeding tube, use an ENFit syringe and remove the plunger from the barrel.
9. Bend or crimp the feeding tube just below the connection to the tip of the irrigation syringe **to prevent the flow of formula as you fill the syringe.**
10. Carefully pour the formula into the syringe barrel.
11. Release the bent or crimped tube, allowing formula to begin to flow. Do not raise the syringe more than 15 to 18 inches above the tube insertion site, **which can cause discomfort because of excessive pressure and too rapid instillation.** Recrimp the tube before the syringe runs completely dry **to prevent air from filling the line and infusing, as added formula pushes the air in front of it.**
12. When the syringe is nearly empty, refill it in the same manner until the feeding is completed. Monitor the patient throughout the feeding, **observing for any signs of respiratory distress.**
13. Flush the tube with 50 to 100 mL of water to clear formula from the line and ensure that the tube is patent. **This will reduce clogging and bacterial growth.**
14. Administer water via the feeding tube between feedings or as ordered.
15. Maintain the patient in a semi-Fowler's position or with the head of the bed elevated at least 30 degrees for 1 hour **to promote gastric emptying and prevent vomiting or aspiration.**
16. Follow the Ending Implementation Steps located on the inside back cover.

Evaluation Steps

1. Observe and assess the patient's respiratory status post-feeding because changes such as increased rate, wheezing, increased coughing, or sudden decrease in oxygen saturation **can indicate aspiration.**
2. Auscultate breath sounds at least once every 8 hours **to detect adventitious sounds that may indicate aspiration.**
3. Measure aspirate every 4 hours **to evaluate the patient's tolerance of feedings.**
4. Monitor laboratory values, including albumin, prealbumin, and transferrin. Notify the health-care provider of significant changes. **This provides information to determine whether the patient's nutritional status is stable, improving, or worsening.**
5. Monitor intake and output every shift and calculate 24-hour totals, **noting if there is an imbalance.**

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5. Monitor intake and output every shift and calculate 24-hour totals, **noting if there is an imbalance.**
6. Weigh the patient daily, at the same time of day and with the same scale to maintain consistency, until the tube feeding maximum rate has been reached; then weigh three times per week or per the health-care provider's order, noting if weight begins to decline. **Evaluate the effectiveness of feedings.**
7. For patients with tubes placed through the abdominal wall, inspect the insertion site for erythema, edema, increased warmth, and drainage **to detect signs and symptoms of infection.**

Sample Documentation

06/02/28 0645 Awake and alert, lying supine. NG tube placement checked by aspirating residual gastric contents, obtained 75 mL of yellowish-brown gastric contents, pH - 3. Insertion mark remains at point of entry. C/O of mild discomfort in left nostril, no erythema noted, skin intact. Replaced tape securing NG tube to nose. States it feels better. Head of bed elevated 45 degrees. Tube flushed with 50 mL of water, and 250 mL of Jevity instilled per gravity. Tube flushed with 50 mL water. No signs of gastric or respiratory distress noted. _____

Nurse's signature and credentials

Skill 24.7 Using a Feeding Pump

Assessment Steps

1. Review the patient's medical record and health-care provider's order to determine the rate, route, and type of formula to be administered **to reduce the risk of error.**
2. Evaluate the patient's current nutritional status. Review the laboratory data and other available diagnostic data, such as fingerstick blood sugar readings. Assess for fluid volume excess or deficit and metabolic or electrolyte imbalances.
3. Assess whether the patient has impaired swallowing or has had facial trauma or surgeries of the alimentary canal, head, or neck. **These conditions indicate the patient's need for enteral tube feedings.**
4. Obtain a baseline weight to determine increases or decreases after initiation of feedings. **This provides for later comparisons to help evaluate effectiveness of feedings.**
5. For PEG tubes or J-tubes, inspect the skin around the tube for irritation, signs of skin breakdown, infection, or drainage.
6. Auscultate bowel sounds before feeding. If no bowel sounds are heard, consult the health-care provider before initiating feeding.

Planning Steps

1. Gather needed equipment and supplies: clean gloves, stethoscope, pH strips, tape measure, prescribed enteral formula, disposable feeding bag and tubing or a ready-to-hang system with ENFit connections, ENFit syringe for small-bore feeding tube, and feeding pump. **Being prepared saves time and effort and increases efficiency. Safety: Note that a feeding pump is not the same as an IV pump.**
2. Plan to change the bag every 24 hours or according to facility policy. Bags should be flagged with the date and time they are hung **so it is easy to tell when to change them.**

Implementation Steps

1. Follow the Initial Implementation Steps located on the inside back cover.
2. Have the tube feeding formula at room temperature **to reduce intestinal cramping.**
3. Check the expiration date of the formula and the integrity of its container **to prevent using outdated or spoiled formula.**
4. Prepare a ready-to-hang system or pour formula into a disposable feeding bag with clamped tubing. Hang the formula bag on the IV pole.
5. Place the patient in the high Fowler's position or elevate the head of the bed 30 to 45 degrees **to reduce the risk of reflux and aspiration.**
6. Prime the tubing from the formula bag by opening the clamp and allowing formula to flow until all air in the tubing is displaced with formula. **Instillation of air can cause discomfort, belching, and reflux.**
7. Position a waterproof pad under the distal end of the NG tube, PEG tube, or J-tube **to protect the bed from soiling.**
8. Put on clean gloves and proceed to verify tube placement. Assess the location of the insertion mark to determine that it is still at the point of entry. Measure the length of the tube extending from the insertion site to the distal end of the tube. Use an irrigation syringe to aspirate residual gastric contents. If your facility uses pH to check placement, ensure that it is between 1 and 4. **This reduces the risk of instillation into a displaced tube.**

Insertion into a Displaced Tube.

9. Return the entire aspirate to the stomach **to prevent electrolyte imbalance**. If the residual is 250 mL or greater and the patient has no symptoms of abdominal pain or a distended abdomen and no nausea or vomiting, withhold the feeding and reassess the residual in 1 hour. Once the residual volume is within acceptable parameters, you may proceed with the feeding process.
10. Flush the tube with 50 to 100 mL of water **to ensure that the tube is clear and patent**.
11. Remove gloves and connect the feeding setup tubing to the patient's feeding tube, thread it through the feeding pump, set the volume and rate, and initiate the feeding.



12. Maintain the patient in the semi-Fowler's position or with the head of the bed elevated continuously **to promote gastric emptying and prevent vomiting or aspiration**.
13. Follow the Ending Implementation Steps located on the inside back cover.

Evaluation Steps

1. Measure aspirate every 4 hours **to evaluate the patient's tolerance of feedings**.
2. Monitor all laboratory values regularly: Albumin, prealbumin, and transferrin indicate stable, improving, or worsening nutritional status. Notify the health-care provider of significant changes. **This evaluates nutritional status and effectiveness of feedings.**
3. Observe and assess the patient's respiratory status post-feeding because changes such as increased rate, wheezing, increased coughing, or sudden decrease in oxygen saturation **can indicate aspiration**.
4. Monitor intake and output every shift and calculate 24-hour totals, **noting imbalance**.
5. Weigh the patient daily, at the same time of day and with same scale to maintain consistency, for the first 2 weeks of continuous feeding, noting any decrease. Then weigh three times per week or per the health-care provider's order, noting if weight begins to decline. **This evaluates the effectiveness of feedings.**

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Sample Documentation

03/24/28 1130 NG tube placement verified by x-ray. Jevity 500-mL tube feeding initiated per feeding pump 50 mL/hr. Hob elevated 45 degrees. _____ Nurse's signature and credentials

CRITICAL THINKING CONNECTIONS: POST CONFERENCE

After caring for your patient for 2 days, you feel much more comfortable around NG tubes. You inserted his tube on the first day, and placement was verified by aspirating greenish stomach contents, which had a pH of 3, and also by x-ray before instilling anything into the tube. You were still nervous, though, when you first attached the feeding pump and started the feedings. You kept watching the patient and checking for any signs of aspiration. On the second day, the patient pulled out his tube, and you had to insert a new one. This time it went even more smoothly, and you began to feel confident about this skill. Now you are glad that you were the student assigned to this patient.

Key Points

- Methods for assisting with meals for patients in the hospital include preparing the environment and preparing the patient in addition to assisting the patient with eating, if needed.
- It is important to carefully measure and record all fluids taken in as intake, whether they are taken in by mouth, are given via IV, or are administered as enteral or parenteral fluids. It is also important to measure and record all bodily fluids that are lost as output, including urine, emesis, liquid stool, blood, gastric contents, and drainage. These measures give valuable information about the patient's fluid balance.
- NPO status means *nothing by mouth* and is usually ordered for a limited time as preparation for procedures or tests. Regular diets are ordered for patients without special nutritional needs. Diets that are modified by consistency include clear liquid, full liquid, mechanical soft, and pureed.
- Some common diets modified for disease are diabetic, sodium-restricted, fat-restricted, and renal diets. Diets that may be modified by preference include gluten-free and vegan.
- Nurses are responsible for monitoring the type of diet, percentage eaten, and whether or not the patient tolerated the meal. They are also responsible for advancing the patient's diet when "Advance as tolerated" is ordered. Patients with diabetes have many special needs, and the nurse is responsible for teaching and assessing the patient for complications such as hypoglycemia and hyperglycemia.
- Common eating disorders include anorexia nervosa, bulimia nervosa, and binge eating disorder.
- Some drugs can affect appetite or digestion because of side effects. Some drugs can alter the absorption, metabolism, and excretion of nutrients by affecting GI and liver function. Some drugs affect the kidneys and interfere with the excretion of nutrients.
- NG tubes are shorter than NI tubes because they go only to the stomach. NI tubes are usually smaller in diameter and more flexible than NG tubes. The double-lumen tube, also called a Salem sump, is used strictly for gastric decompression.
- Gastric decompression is the process of reducing the pressure within the stomach by emptying it of its contents. An NG tube is inserted into the stomach and connected to low intermittent suction to accomplish this.
- Patients with difficulties tolerating oral intake may receive enteral feedings. Tube feedings can be NG or NI or via the lower GI tract via a PEG tube or J-tube.
- Small-bore feeding tubes require the use of ENFit tubes, bags with tubing, and syringes. These will only connect to one another and will not connect to IV tubing to prevent misconnection errors.
- Tube feedings can be continuous or intermittent bolus. Continuous feedings should be administered with a feeding pump.
- Verification of NG/NI tube placement by radiography is the most accurate method. Aspiration of gastric contents and assessment of the pH are used to verify continued presence of the tube in the proper location of the GI tract.
- Complications can arise with enteral tube feeding, such as clogged tubes, aspiration, electrolyte

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- Complications can arise with enteral tube feeding, such as clogged tubes, aspiration, electrolyte imbalance, hyperglycemia, and severe diarrhea.
- There are two types of parenteral nutrition: PPN and TPN. PPN solutions contain a lower concentration of dextrose than TPN. When administering any type of parenteral nutrition, the nurse must adhere strictly to sterile technique; contamination can lead to serious infection and septicemia.

CRITICAL THINKING EXERCISES

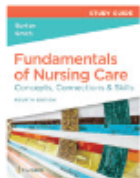
Answers available online.

1. You are working on a cardiovascular unit and are assigned to care for a 68-year-old female who is being admitted. She has advanced multiple sclerosis, virtually no motor function from the neck down, and significant dysphagia. What are your nursing priorities for this patient? Develop a plan of care to assist her in meeting her nutritional needs.
2. A 75-year-old male has been ill with nausea, vomiting, and diarrhea for the last 4 days. He transferred from his long-term care (LTC) facility to your hospital. His weight 1 week ago at the LTC was 165 lb (75 kg); today it is 156 lb (70.9 kg). Your assessment reveals hyperactive bowel sounds, a soft and slightly tender abdomen in all four quadrants, and continued nausea and diarrhea but no vomiting in the last 8 hours. He is afebrile with normal electrolyte values. What type of diet order would be the best choice for him? What nursing interventions should you perform during your 7-to-3 shift?
3. You are caring for a patient receiving 250 mL of formula via intermittent enteral feedings. You withdraw 50 mL of residual gastric content when you check his NG tube's placement, patency, and residual. Should you return the gastric aspirate to the patient's stomach or discard it? Provide a rationale for your response.

Additional Resources



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Study Guide

